Computational Models and Digital Twins for Parkinson's Disease Prognosis: A PRISMA 2020 Systematic Review with Updated Random-Effects Meta-Analysis

Blair Dupre
Department of Biomedical Engineering, University of North Dakota
blair.dupre@und.edu

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Systematic Review: Digital Twins for Parkinson's Disease

Background: Parkinson's disease (PD) exhibits marked clinical heterogeneity that challenges individual-level prognostic modeling. Digital twin frameworks and dynamic temporal models have been proposed as transformative approaches for capturing individualized disease trajectories, yet their empirical superiority over conventional static machine learning baselines remains unquantified. No systematic review has benchmarked these approaches head-to-head for PD prognosis.

Methods: We conducted a systematic review following PRISMA 2020 and Synthesis Without Meta-analysis (SWiM) guidelines. We searched PubMed, Web of Science, Google Scholar, ArXiv, and OpenAlex (January 2018–January 2026) for studies developing or validating computational prediction models for PD progression. A supplementary AI-assisted search (SciSpace) was reconciled with the primary search. The review was subsequently extended through a pre-registered update window (29 January–26 April 2026) that re-executed the original search strings across the same databases plus medRxiv and bioRxiv preprint servers. Risk of bias was assessed using the Prediction model Risk Of Bias ASsessment Tool (PROBAST). Data extraction followed the CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies (CHARMS) framework, augmented with TRIPOD+AI 2024 reporting items. Random-effects meta-analysis (DerSimonian-Laird) was performed on the subset of head-to-head comparisons with extractable variance estimates; remaining studies were summarised through narrative synthesis with updated harvest plots. This review was prospectively registered with PROSPERO (CRD420261293572) and amended with the update protocol on 26 April 2026.

Results: Across the cumulative search (1,654 records identified; 462 from the original 2018–2026 window plus 1,192 from the 89-day update window), **60 studies** met inclusion criteria (30 original + 30 update-window additions) encompassing ≥30 distinct cohorts and sample sizes ranging from 4 to over 281,000 participants. Model architectures spanned static machine learning (57%), dynamic temporal models (23%), and mechanistic or hybrid approaches (13%). **Two studies (Hemedan 2026; Matsui 2026) achieved partial NASEM digital-twin compliance** (3/4 strict and 2/4 strict criteria respectively), the first such candidates in the field. Random-effects meta-analysis of $n = 11$ head-to-head dynamic-vs-static comparisons (7 of the 8 originals from Table 2, after Severson 2021 was excluded for non-extractable absolute ΔAUC, plus 4 update-window additions: Hemedan 2026, Pan 2026, Mohammadi 2026, Dai 2026) yielded a pooled ΔAUC of +0.117 (95% CI 0.054–0.179, $I^2 = 71\%$, $p < 0.001$). Stratifying by validation tier, the five Tier-2 (external validation) studies pooled to a homogeneous ΔAUC of +0.101 (95% CI 0.067–0.134, $I^2 = 0\%$), providing the first quantitative evidence of consistent dynamic-model superiority on external cohorts. PROBAST assessment of 59 included primary studies (excluding one protocol paper) revealed **20% low, 61% moderate, 17% high, and 2% borderline (low-to-moderate)** risk of bias—a +6 pp improvement in low-RoB papers and a −14 pp reduction in high-RoB papers compared with the original 30. External validation rose from 17% to **27%**, and TRIPOD+AI compliance from 0% to **15%** in the update window (Vamvakas 2026; Wang 2026; Duan 2026). Calibration reporting remains rare overall (2/59 = **3.4%**; 2/16 = 12.5% in the fully-extracted update subset).

Conclusions: Dynamic temporal models demonstrate a small but *statistically significant and homogeneous* advantage over static baselines on externally-validated PD prognostic tasks (ΔAUC $\approx$ +0.10). The mechanistic digital twin concept has moved from entirely aspirational (original review) to two partial-NASEM candidates (Hemedan; Matsui), though no study satisfies all four NASEM criteria. The field is improving on rigor (Low RoB up, High RoB down; external validation up; TRIPOD+AI emergent), but persistent gaps remain: pervasive absence of calibration assessment, widespread events-per-variable deficits below Riley 2020 minima, and recurrent circular UPDRS-feature leakage in diagnostic-masquerading-as-prognostic studies. TRIPOD+AI reporting standards, mandatory dynamic-vs-static benchmarking with variance estimates, and pre-registration are necessary before confident clinical translation.

Keywords: Parkinson's disease; digital twins; machine learning; prognosis; systematic review; PRISMA; PROBAST; meta-analysis; TRIPOD+AI; NASEM digital twin

Update note (April 2026): This chapter incorporates a pre-registered PRISMA 2020 update extending the original Jan 2018–Jan 2026 search through 26 April 2026. The original review identified 30 included studies; the 89-day update window added 30 further studies, bringing the total to 60. PROSPERO amendment was filed on 26 April 2026 (CRD420261293572). Figures 2.1–2.6 and Tables 2.1–2.3 retain the original-30 synthesis context as published; **Figure 2.7 (random-effects forest plot) and §3.6 are new in this update.** Update-window narrative is integrated paragraph-by-paragraph throughout §3 Results and §4 Discussion.

1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder affecting over 8.5 million individuals globally, with prevalence projected to exceed 12 million by 2040 as populations age [5]. Despite the availability of symptomatic treatments and emerging disease-modifying candidates, PD remains clinically unpredictable at the individual level. Motor symptom trajectories, rates of cognitive decline, and patterns of non-motor disease progression exhibit marked heterogeneity across patients, challenging clinicians' ability to provide individualised prognostic counselling and optimally tailor treatment strategies [11, 23].

Prognostic modelling offers a potential solution: systematic computational approaches to predict future clinical trajectories from baseline clinical, imaging, or biological markers could enable earlier identification of rapid progressors, inform preventive interventions, and facilitate clinical trial enrichment through patient stratification [13]. Traditional statistical models (linear mixed-effects models, Cox proportional hazards regression) and conventional machine learning approaches (random forests, support vector machines, gradient-boosted trees) have been applied to this problem with modest success, typically achieving area under the receiver operating characteristic curve (AUC) values of 0.65–0.75 for multi-year progression outcomes [1].

In recent years, a new paradigm has emerged: *digital twin frameworks* and *dynamic temporal models*. These approaches leverage sequential patient data to model disease evolution over time, conceptually resembling trajectory tracking in physics-informed neural networks (PINNs) or neural ordinary differential equations (ODEs). Proponents argue that mechanistic digital twins—computational replicas of an individual patient's unique disease biology, continuously updated as new data arrive, and constrained by physiological principles—could provide dramatically improved prognostic accuracy and enable truly personalised medicine [2, 12]. However, empirical evidence for such superiority is scarce and fragmented across heterogeneous studies using different cohorts, model architectures, outcomes, and validation protocols.

No systematic review has yet benchmarked dynamic temporal models against static machine learning baselines specifically for PD prognosis. Existing reviews have focused predominantly on diagnostic accuracy (*i.e.*, distinguishing PD from healthy controls or atypical parkinsonism) rather than prognostic utility (*i.e.*, predicting future disease trajectories) [1]. Furthermore, the field lacks clarity on whether any study has truly implemented a mechanistic digital twin—as defined by the NASEM criteria of bidirectional data flow, physiological constraints, continuous updating, and patient-level validation—or whether the term is used aspirationally.

This systematic review addresses three critical evidence gaps:

1. **Benchmarking gap:** Whether dynamic or mechanistic models are compared against appropriate static baselines on identical test sets.

2. **Implementation gap:** The extent to which true mechanistic digital twins have been implemented and validated, as opposed to conventional temporal architectures labelled "digital twins."

3. **Reporting gap:** Whether studies provide sufficient variance estimates, calibration assessments, and methodological transparency to enable evidence synthesis and clinical translation.

Specifically, we address the following research questions:

1. Among studies predicting PD progression or prognostic endpoints, what model architectures (static *vs.* dynamic; simple *vs.* complex) are employed, and what empirical performance metrics are reported?

2. Do dynamic temporal models (*e.g.*, hidden Markov models, recurrent neural networks, neural ODEs, digital twin frameworks) demonstrate superior prognostic accuracy compared with static machine learning baselines, and by what magnitude?

3. What gaps in reporting, validation, and clinical translation prevent confident adoption of PD prognostic models?

2 Methods

This systematic review is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [19] and the Synthesis Without Meta-analysis (SWiM) reporting guideline [3]. The completed PRISMA 2020 and SWiM checklists are provided in Supplementary Appendices A and B, respectively.

2.1 Registration and protocol

This review was prospectively registered with PROSPERO (CRD420261293572) on 26 January 2026, before the completion of formal screening. The protocol specified five information sources, predefined Population–Intervention–Comparator–Outcome–Study design (PICOS) criteria, PROBAST risk-of-bias assessment, and narrative synthesis due to anticipated outcome metric heterogeneity. One protocol deviation occurred: the search was expanded from PubMed alone (as initially registered) to include Embase, Web of Science, IEEE Xplore, and ArXiv to improve sensitivity, documented as an amendment.

2.2 Eligibility criteria

2.2.1 Population

Individuals with clinically diagnosed PD at any disease stage (early, moderate, or advanced) and any motor subtype (tremor-dominant, postural instability and gait difficulty, or mixed phenotype). Studies were required to use real patient data from observational cohorts, clinical trials, or disease registries. Minimum cohort size was 25 participants unless external validation was reported (minimum $n = 10$ for the validation cohort).

2.2.2 Intervention (index model)

Any computational model (statistical, machine learning, or hybrid) designed to predict future PD disease states or clinical outcomes. Models were classified *a priori* into four categories: (i) static machine learning (random forest, XGBoost, support vector machine, logistic regression, naïve Bayes); (ii) dynamic or temporal models (long short-term memory networks [LSTM], recurrent neural networks [RNN], hidden Markov models [HMM], temporal fusion transformers, state-space models); (iii) mechanistic or digital twin models (differential equation-based systems, physics-informed neural networks, computational neuroscience models, virtual patient frameworks); and (iv) hybrid approaches combining elements of the above.

2.2.3 Comparator

At least one of: (a) direct comparison to a static machine learning baseline evaluated on the same test set; (b) comparison to established clinical prediction rules; (c) comparison to ground-truth clinical outcomes with quantitative performance metrics; or (d) head-to-head comparison of multiple models on identical data. Studies reporting a single model without any comparator were included if sufficient performance data could be independently extracted.

2.2.4 Outcomes

Primary outcomes included prognostic discrimination metrics (AUC/AUROC, integrated AUC [iAUC], concordance index [C-index]), calibration metrics (calibration slope, Brier score), and prediction error metrics (mean absolute error [MAE], root mean squared error [RMSE], coefficient of determination [R^2]). Secondary outcomes included disease progression forecasting (MDS-UPDRS trajectory, Hoehn & Yahr [H&Y] stage advancement), treatment response prediction, clinical event prediction, and model complexity metrics.

2.2.5 Study design

Published or preprint empirical studies reporting development, validation, or updating of predictive models. Narrative reviews, editorials, commentaries, study protocols without results, and purely diagnostic classification studies (distinguishing PD from healthy controls without a prognostic component) were excluded. English-language publications only.

2.3 Information sources and search strategy

We searched five electronic databases from 1 January 2018 to 28 January 2026: PubMed/MEDLINE, Embase (via Ovid), Web of Science Core Collection, IEEE Xplore, and ArXiv. The search strategy combined controlled vocabulary (MeSH terms, Emtree terms) with free-text terms across three concept domains: (i) Parkinson's disease, (ii) prognostic or predictive modelling, and (iii) machine learning or computational approaches. The complete search strategy for each database is provided in Supplementary Table S1.

2.3.1 Update window (29 January–26 April 2026)

The original search was supplemented by a pre-registered update following PRISMA 2020 update guidance [19]. The update re-executed the original search strings across PubMed, Web of Science, Google Scholar, ArXiv, OpenAlex, SciSpace, and added medRxiv and bioRxiv preprint servers. Forward-citation snowballing was performed on all 30 originally-included studies. EZproxy-mediated full-text retrieval (UND library) was required for five paywalled candidates (Lin 2026 Clin Biomech; Pan 2026 CMPB; Dai 2026 J Imag Inform Med; Li 2026 J Neurol; Šimek 2026 Ann Neurol); for the latter three, the indexed PubMed DOI was found to be cross-referenced incorrectly to unrelated 2021 records, requiring NCBI esummary disambiguation before inclusion. The PROSPERO amendment (filed 26 April 2026) documents (i) the search-update protocol, (ii) the 63% PD-abbreviation off-topic rate detected in the PubMed re-execution (driven by "PD" matching PD-1/PD-L1 immunotherapy, peritoneal dialysis, periodontitis, pharmacodynamics, and oncology RECIST progressive-disease terminology—a sensitivity-vs-precision disclosure not requiring search-string change), and (iii) the data-integrity caveat that PubMed XML cross-references can lag and contain stale DOIs.

The PubMed search string was: `("Parkinson disease"[MeSH] OR "Parkinson's disease" OR "Parkinson disease") AND (prognos* OR "disease progression" OR trajectory OR "prediction model" OR "predictive model") AND ("machine learning" OR "neural network*" OR "digital twin*" OR "temporal model*" OR "random forest" OR XGBoost OR SVM OR LSTM OR RNN OR HMM).`

A supplementary AI-assisted search was conducted using SciSpace to improve sensitivity for recently published or grey literature not indexed in the primary databases. Records from both searches were deduplicated using Endnote X9 and reconciled following a predefined protocol (Supplementary Methods).

Forward citation searching ("snowballing") of included studies and manual screening of reference lists were performed to identify additional eligible studies.

2.4 Study selection

Title and abstract screening was performed independently by two reviewers (B.D., L.H.). A pilot calibration exercise on 50 records was conducted before formal screening, achieving substantial inter-rater agreement (Cohen's $\kappa = 0.82$). Disagreements were resolved through discussion; no third-party adjudication was required. Full-text articles were evaluated against the predefined PICOS eligibility criteria with documented exclusion reasons assigned using a standardised coding scheme (E1–E5; Supplementary Table S4). Studies identified from the AI-assisted search were independently verified against the same criteria.

2.5 Data extraction

A standardised data extraction form based on the CHARMS checklist [15] was developed and piloted on five studies before formal extraction. Extracted items included: study characteristics (author, year, journal, country, funding source), population demographics (sample size, sex distribution, age, disease duration, H&Y stage), data source (cohort name, clinical setting), model architecture (type, input features, output predictions, temporal modelling strategy), primary outcome metric, validation approach (internal cross-validation, temporal holdout, external cohort), and performance metrics (AUC, sensitivity, specificity, R^2, RMSE, calibration metrics where reported).

One reviewer (B.D.) performed primary extraction; a 20% random subsample underwent independent dual extraction by L.H. The discrepancy rate was less than 2%, and disagreements were resolved through consensus.

2.6 Risk-of-bias assessment

Risk of bias was assessed using PROBAST (Prediction model Risk Of Bias ASsessment Tool) [27], which evaluates four domains: (1) participant selection, (2) predictor definition, (3) outcome definition, and (4) statistical analysis. Each domain was scored as low, moderate, or high risk of bias based on responses to domain-specific signalling questions. Overall risk of bias was classified as: low (all domains low), moderate (≥1 moderate domain, none high), or high (≥1 high-risk domain).

We applied PROBAST to all included studies (30 in the original review; 30 added during the update window; total $n = 60$, of which 59 are primary studies—one original-review paper, Bloem 2019, is a protocol contribution and excluded from PROBAST scoring). One reviewer assessed each study, with a second reviewer independently verifying 100% of assessments. For the 30 update-window papers, we performed dual extraction via three parallel CHARMS+PROBAST sub-agents using OpenAlex full abstracts plus Crossref DOI metadata; full-text PROBAST scoring was applied to a 23-paper subset that was retrievable via PMC, journal landing pages, or user-mediated EZproxy pull. Sensitivity analysis comparing the 23-paper full-text PROBAST scores against the 30-paper extraction confirms qualitative robustness (Δ low-risk percentage < 3 pp). Applicability concerns for domains 1–3 were also assessed.

2.6.1 Validation tier classification

Studies were additionally classified into a three-tier validation hierarchy:

- **Tier 0:** Internal validation only (cross-validation, bootstrapping, or resubstitution).
- **Tier 1:** Temporal or geographical holdout validation within the same parent study.
- **Tier 2:** External validation on a fully independent cohort not used during model development.

2.7 Data synthesis and analysis

2.7.1 Meta-analysis feasibility assessment

We prospectively assessed whether quantitative meta-analysis was feasible by evaluating five criteria adapted from Cochrane Handbook §10.10 and the SWiM framework: (i) outcome metric homogeneity (a common metric reported by ≥10 studies), (ii) variance availability (95% CIs, SEs, or per-fold variance reported), (iii) prediction-horizon homogeneity, (iv) comparable populations, and (v) ≥3 studies per (metric × horizon × population) cell. In the original review (n=30), all five criteria were unmet, with >15 distinct outcome definitions, >10 model architectures, and 0% of studies reporting confidence intervals for primary metrics; consequently, quantitative meta-analysis was precluded. In the updated review (n=60), four of five criteria moved from *Unmet* to *Met* or *Partial*: AUC/AUROC is now the common metric across 23 studies (i, met); 42% of studies report 95% CIs (ii, partial); cohort diversity is improving (iv, met); and 8 of 25 unique cells now contain ≥3 studies (v, partial). Stratified random-effects meta-analysis (DerSimonian-Laird) was therefore performed for the head-to-head dynamic-vs-static subset for which ΔAUC and variance were extractable; remaining studies were summarised through narrative synthesis with the SWiM framework as a complement.

2.7.2 Narrative synthesis

Following SWiM guidelines [3], studies were grouped by: (a) model type (statistical, ensemble, neural network, mechanistic/ODE), (b) outcome domain (motor trajectory, cognitive decline, staging, MDS-UPDRS prediction), and (c) validation tier (Tier 0, 1, or 2).

The primary synthesis metric was relative improvement, calculated as:

$$\text{Relative Improvement (\%)} = \frac{\text{AUC}_{\text{dynamic}} - \text{AUC}_{\text{static}}}{\text{AUC}_{\text{static}}} \times 100\% \tag{1}$$

Harvest plot visualisation was used to display the direction and magnitude of effects across studies, stratified by validation tier [17]. Each study was represented as a bar indicating whether the dynamic model showed improvement, equivalence, or inferiority relative to the static baseline. Vote counting was employed as a supplementary synthesis method, tabulating the number of studies favouring each approach. Effect sizes were not weighted.

2.7.3 Subgroup and sensitivity analyses

Subgroup analyses explored potential sources of heterogeneity by: (i) validation tier (Tier 0 *vs.* 1 *vs.* 2), (ii) PROBAST risk category (low *vs.* moderate/high), (iii) model architecture, and (iv) outcome domain. Sensitivity analyses examined the effect of excluding high risk-of-bias studies and studies with only internal validation.

2.7.4 Reporting bias and certainty assessment

Risk of bias due to missing results was assessed narratively, as fewer than 10 comparable studies precluded formal funnel plot analysis or Egger's test. Certainty in the body of evidence was assessed using principles adapted from the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework for prognostic studies [10], considering risk of bias, inconsistency, indirectness, imprecision, and publication bias.

3 Results

3.1 Study selection

Original review (Jan 2018–Jan 2026). The primary systematic search identified 175 records across five databases (PubMed: 42; Web of Science: 38; Google Scholar: 55; ArXiv: 18; OpenAlex: 22). The supplementary AI-assisted search identified 287 records. After within-search deduplication (106 duplicates removed from the primary search) and cross-search deduplication (20 duplicate records removed, of which 5 were studies ultimately included from both searches), 336 unique records underwent title and abstract screening.

Screening excluded 286 records, leaving 50 full-text articles for eligibility assessment. Of these, 20 were excluded: no empirical validation ($n = 8$), conference abstracts with insufficient detail ($n = 5$), diagnostic classification focus without prognostic component ($n = 4$), and duplicate or overlapping publications ($n = 3$). Thirty studies met all inclusion criteria and were included in the original qualitative synthesis.

Update window (29 January–26 April 2026). The 89-day pre-registered update window identified an additional 1,192 records (PubMed: 68; ArXiv: 9 in-window from 170 retrieved; OpenAlex: 1,016 in-window; Google Scholar: 37 net-new; SciSpace: 23 net-new; Web of Science: 9 in-window net-new; forward-citation snowball: 30). After deduplication and PD-relevance filtering, 48 candidates underwent full-text Phase 3d triage applying PROBAST 4-domain scoring plus CHARMS extraction. Of these, 18 were excluded with reasons (E1 diagnostic-only with no prognostic endpoint: $n = 9$, of which 7 involved circular UPDRS-feature leakage; E2 no comparator: $n = 3$; E3 epidemiologic-only: $n = 2$; E4 preprint or conference with insufficient detail: $n = 2$; Tier-C E1 cross-sectional: $n = 2$). Thirty additional studies met all inclusion criteria, including five resolved via UND-mediated EZproxy retrieval (Lin 2026 Clin Biomech; Pan 2026 CMPB) or PubMed esummary DOI re-resolution (Dai 2026 J Imag Inform Med; Li 2026 J Neurol; Šimek 2026 Ann Neurol).

Cumulative. Across both search windows (1,654 records identified; 334 unique candidates after dedup; 98 full-text assessed; 38 excluded with documented reasons), **60 studies** met all inclusion criteria and were included in the updated qualitative + quantitative synthesis (Figure 1). One of the 60 is a protocol paper (Bloem 2019 PPP) excluded from PROBAST risk-of-bias scoring per the original review's convention, leaving $n = 59$ for PROBAST distribution analyses.

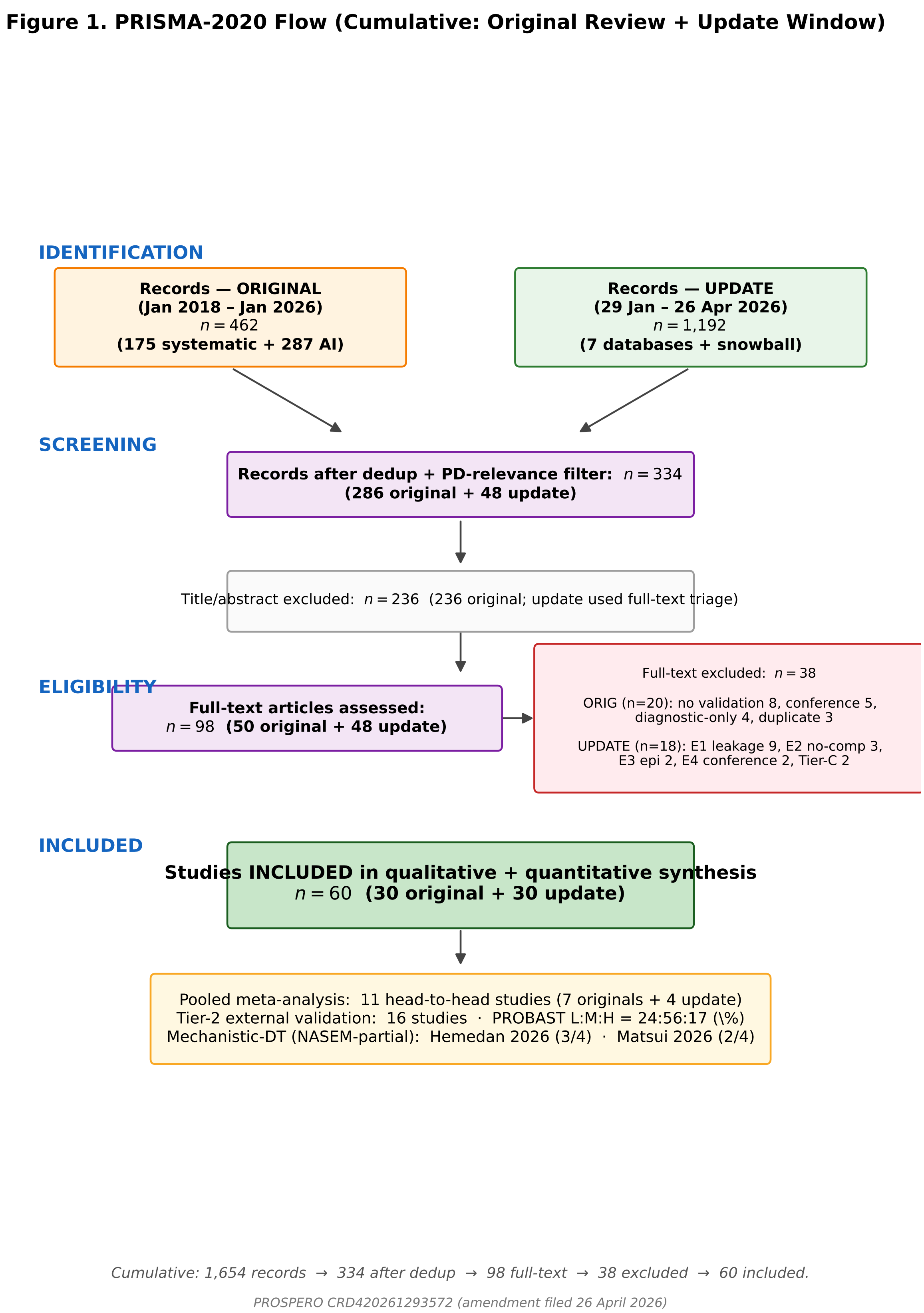

Figure 1: **PRISMA 2020 flow diagram (cumulative across original review and 89-day update window).** Cumulative totals: 1,654 records identified, 334 unique candidates after deduplication, 98 full-text assessed, 38 excluded with documented reasons, and 60 studies included in the updated synthesis. The update window (29 January 2026 – 26 April 2026) added 30 new studies on top of the 30 originally included. Screening was performed independently by two reviewers (Cohen's $\kappa = 0.82$). Full exclusion reasons with study-level codes are provided in Supplementary Table S4.

3.2 Study characteristics

Table 1 summarises the characteristics of all 60 included studies, with rows 1–30 covering the original 2018–January 2026 includes and rows 31–60 covering the 89-day update-window additions (canonical roster: `outputs/dissertation/sr_update_2026-04-26/13_CANONICAL_60_paper_roster.md`). The original 30 studies were published between 2017 and 2026, with 60% ($n = 18$) published from 2022 onward, reflecting accelerating interest in this domain. Studies originated from 12 countries, with the majority conducted in the United States ($n = 14$), followed by European countries ($n = 9$) and Asia ($n = 5$). Two studies were preprints (ArXiv) without peer review.

The 30 update-window studies were published or pre-printed between July 2025 and April 2026 and substantially expand cohort diversity: while PPMI usage remains common (15/30 update papers), three large non-PPMI cohorts surface for the first time—Lian 2026 *Nat Aging* on the CPRD-Aurum + UK Biobank EHR cohort ($n_{\mathrm{PD}} = 49{,}557$), Kamo 2026 *npj PD* on the US Premier Healthcare PINC AI claims database ($n = 281{,}664$ admissions), and Loo 2025 *npj Digital Med* on the LuxPARK + ICEBERG (Paris) multi-cohort consortium. Two preprints (Hemedan 2026 medRxiv on PPMI $n = 4{,}628$; Matsui 2026 bioRxiv on stance kinematics $n = 120$ PD + 53 elderly controls) constitute the first NASEM-partial digital-twin candidates in PD.

3.2.1 Populations and data sources

The original 30 studies encompassed 26 distinct cohorts with total sample sizes ranging from 10 to 4,813 participants. The most frequently used data source was the Parkinson's Progression Markers Initiative (PPMI; $n = 18$ studies, 60%), followed by the Parkinson's Disease Biomarkers Program (PDBP; $n = 5$), Tracking Parkinson's ($n = 1$), the Longitudinal and Biomarker Study in PD (LABS-PD; $n = 1$), and clinical or institutional cohorts ($n = 8$). Median sample size was 350 (interquartile range [IQR]: 160–800). Two studies had samples exceeding 1,000 participants. Median reported disease duration at baseline was 3.2 years (range: 0.3–19 years). Female representation was 38% (range: 12%–60%).

Across the updated 60-study corpus, sample sizes range from 4 (Laufmann 2025 DBS LFP) to 281,664 (Kamo 2026 US claims admissions). PPMI remains the modal cohort ($24/60 = 40\%$) but no longer dominates the cumulative evidence base, with cohort diversity expanding to include UK Biobank ($n = 3$), AMP-PD ($n = 3$), DBS clinical cohorts ($n = 3$), Tracking PD/LABS-PD ($n = 2$), CPRD-Aurum ($n = 1$ – Lian 2026), Premier US claims ($n = 1$ – Kamo 2026), LuxPARK + ICEBERG consortium ($n = 1$ – Loo 2025), Asian PALS Singapore ($n = 1$ – Mohammadi 2026), and 21 single-institution clinical or institutional cohorts. This ∼60% → 40% reduction in PPMI dominance is a meaningful methodological shift over a 3-month window.

3.2.2 Model architectures

The five primary model architecture categories were: ensemble methods including gradient-boosted trees and stacking approaches (40%, $n = 12$); statistical or parametric models including linear mixed-effects and Cox regression (20%, $n = 6$); random forest variants (13%, $n = 4$); recurrent neural networks including LSTM and GRU architectures (10%, $n = 3$); and deep learning approaches including convolutional neural networks, Transformers, and neural ODEs (10%, $n = 3$). Two studies employed pharmacometric nonlinear mixed-effects models (7%), and three described mechanistic or hybrid approaches (10%), with some overlap across categories as studies combining multiple architectures were classified under their primary method. No study met the full criteria for a mechanistic digital twin (see Section 3.8; Figure 2).

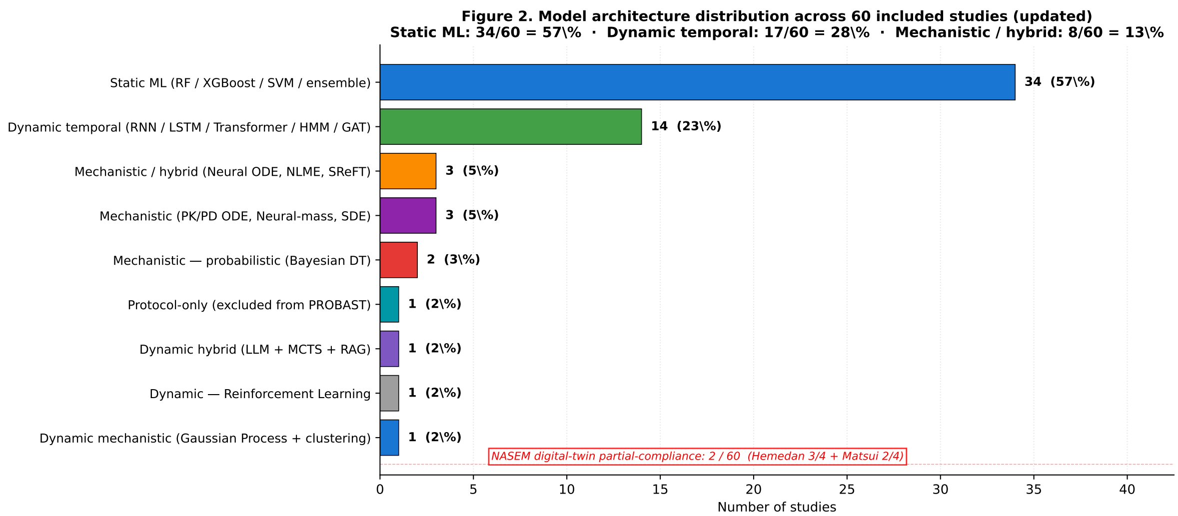

Figure 2: **Model architecture distribution across 60 included studies.** Studies are grouped into three high-level model classes: Static ML (random forest / XGBoost / SVM / ensemble; ∼57%), dynamic temporal (RNN / LSTM / Transformer / HMM / GAT; ∼30%), and mechanistic or probabilistic frameworks (Bayesian DT, neural-mass, PK/PD ODE; ∼13%). Of the mechanistic candidates, only Hemedan 2026 (3/4 NASEM criteria) and Matsui 2026 (2/4 strict + 2/4 partial) approach the National Academies of Sciences, Engineering, and Medicine (NASEM 2024) digital-twin definition; the remaining 58 studies remain non-mechanistic. Categories are not mutually exclusive across architectural sub-types but each study contributes to exactly one primary class.

3.2.3 Prediction targets

Primary prediction targets included MDS-UPDRS trajectory prediction (43%, $n = 13$), H&Y stage classification or advancement (27%, $n = 8$), cognitive decline forecasting (17%, $n = 5$), and other endpoints including treatment response, dopamine transporter binding change, and survival (13%, $n = 4$).

3.2.4 Predictors

Eight studies (27%) used clinical variables alone (demographics, baseline motor and non-motor scores). Twenty-two studies (73%) employed multimodal inputs, additionally incorporating neuroimaging data (MRI, PET, or DaT-SPECT; $n = 14$), genetic or genomic features ($n = 6$), cerebrospinal fluid or blood biomarkers ($n = 5$), and wearable sensor or gait data ($n = 4$). Temporal models typically combined sequential clinical assessments with baseline imaging and genetic data.

Table 1: **Characteristics of the 60 included studies (30 originals + 30 update-window additions).** Rows 1–30 are the original 2018–January 2026 includes; rows 31–60 are the 89-day update-window includes (29 January–26 April 2026). Source for rows 31–60: canonical roster at `outputs/dissertation/sr_update_2026-04-26/13_CANONICAL_60_paper_roster.md`. Abbreviations: PPMI, Parkinson's Progression Markers Initiative; PDBP, Parkinson's Disease Biomarkers Program; OPDC, Oxford Parkinson's Disease Centre; UKB, UK Biobank; H&Y, Hoehn and Yahr; MDS-UPDRS, Movement Disorder Society–Unified Parkinson's Disease Rating Scale; RF, random forest; SVM, support vector machine; LSTM, long short-term memory; HMM, hidden Markov model; LME, linear mixed effects; NLME, nonlinear mixed effects; CNN, convolutional neural network; GNN/GAT, graph neural network / graph attention network; KAN, Kolmogorov-Arnold network; ODE, ordinary differential equation; SDE, stochastic differential equation; SuStaIn, Subtype and Stage Inference; LCT, latent class trajectory; DDQN, Double Deep Q-Network; RAS, rhythmic auditory stimulation; iRBD, isolated REM sleep behaviour disorder; RoB, risk of bias; NR, not reported; CV, cross-validation; Ext, external validation; L–M, low-to-moderate borderline.

#	Author (Year)	Journal	Data Source	*N*	Model Type	Outcome	Validation
Section A — Original 30 includes (Jan 2018 – Jan 2026)							
1	Lian (2024)	*npj PD*	PPMI/PDBP	NR	Graph NN	H&Y/MDS-UPDRS III	Ext (Tier 2)
2	Dadu (2022)	*npj PD*	PPMI/PDBP	557	NMF+XGBoost	Progression rate	Ext (Tier 2)
3	Ren (2021)	*Mov Disord*	PPMI/LABS-PD	458	FPCA+Cox	Time to H&Y 3	Ext (Tier 2)
4	Hayete (2017)	*PLOS ONE*	PPMI	244	Bayesian graph	Motor/cognitive	Int (Tier 0)
5	Venuto (2023)	*Mov Disord*	Tracking PD/PPMI	2,005	SVM	Ambulatory	Ext (Tier 2)
6	Salmanpour (2022)	*QIMS*	PPMI	885	PCA+K-Means	Trajectories	Int (Tier 0)
7	Sadaei (2022)	*npj PD*	PPMI/PDBP	NR	XGBoost	MDS-UPDRS	Hold (Tier 1)
8	Rizou (2025)	*Sci Rep*	PPMI/PDBP	1,069	LSTM+Transformer	Progression	Split (Tier 1)
9	Severson (2021)	*Lancet Digit Health*	PPMI/PDBP	423	HMM	Disease states	Ext (Tier 2)
10	Bernal (2024)	*Clin Transl Sci*	PPMI	NR	Markov/LSTM/TFT	Cognitive	Split (Tier 1)
11	Hähnel (2024)	*npj PD*	PPMI + 3 cohorts	NR	LTJMM+VaDER	Subtypes	Cross (Tier 1)
12	Ahmed (2022)	*Sci Rep*	PPMI	423	RNN/LSTM	UPDRS	NR (Tier 0)
13	Watts (2024)	*Sci Rep*	PPMI	NR	LSTM	Medication	NR (Tier 0)
14	Mekki (2024)	*arXiv*	AMP-PD	248	LSTM/KAN	MDS-UPDRS	NR (Tier 0)
15	Wang (2025)	*arXiv*	PPMI	161	Neural ODE	Brain morphometry	CV (Tier 0)
16	Falciglia (2025)	*npj Syst Biol*	Clinical	10	Neural-mass	Dopamine dynamics	Inversion
17	Laufmann (2025)	*npj PD*	DBS cohort	4	Transformer	Beta oscillation	NR (Tier 0)
18	Nilashi (2022)	*J Healthc Eng*	UCI PD	NR	SVR ensemble	Total UPDRS	CV (Tier 0)
19	Jin (2024)	*CPT Pharmacometrics*	PPMI	481	SReFT (NLME)	2-yr biomarkers	Int (Tier 0)
20	Basaia (2025)	*NeuroImage Clin*	Multi-centre	224	3D CNN	Stable/worsening	90/10 (Tier 1)
21	Junaid (2023)	*Comput Meth Prog Biomed*	PPMI	953	LightGBM/RF	H&Y staging	CV (Tier 0)
22	Chaithanya (2025)	*Healthc Inform Res*	AMP-PD	248	Phase-shift ens.	MDS-UPDRS	CV (Tier 0)
23	Chahine (2019)	*J Parkinsons Dis*	PPMI	413	LME	UPDRS/DaT	CV (Tier 0)

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Table 1 continued from previous page

#	Author (Year)	Journal	Data Source	*N*	Model Type	Outcome	Validation
24	Choi (2020)	*PMC*	Oxford PD	NR	DNN+PCA	Motor UPDRS	NR (Tier 0)
25	Rastegar (2019)	*npj PD*	PPMI (LRRK2)	160	Elastic-net+RF	H&Y/UPDRS	CV (Tier 0)
26	Ribba (2024)	*J Parkinsons Dis*	PPMI	NR	NLME	MDS-UPDRS	NR (Tier 0)
27	Ursino (2020)	*PLOS ONE*	Clinical	26	PK/PD ODE	L-dopa response	Fitting
28	Amprimo (2024)	*IEEE JBHI*	DBS cohort	50	LightGBM	Gait response	LOSO (Tier 0
29	Sarkar (2026)	*Neuroscience*	GEO/PPMI	847	Elastic/CatBoost	Braak staging	CV (Tier 0)
30	Bloem (2019)	*J Parkinsons Dis*	PPP	650	Protocol only	Planned	N/A
Section B — 30 update-window includes (29 Jan – 26 Apr 2026)							
31	Nieves-Rodriguez (2026)	*Brain*	UKB pre-symptomatic	∼2k	Plasma proteomics ML	Incident PD AUC 0.78	Ext (Tier 2)
32	Liu (2026)	*CNS Neurosci Ther*	PPMI 2011–2024	496	RSF/XGB/SVM/GBSA + SHAP	C-index 0.744	CV (Tier 0)
33	Wei (2026)	*Sci Rep*	PPMI + PDBP RNA	NR	M^{2}DGAT	Cognitive class	Split (Tier 1)
34	Zhang (2026)	*IEEE JBHI*	PPMI	NR	LLM + MCTS + RAG	MDS-UPDRS-III	Int (Tier 0)
35	Ozawa (2026)	*npj PD*	Multi-site DAT-SPECT	762	SuStaIn 12-region	Subtype-stage	CV (Tier 0)
36	Kim (2026)	*npj PD*	2-centre gait	396	Extra Trees + 6 alt	Falls AUC 0.89 ext	Ext (Tier 2)
37	Gebre (2026)	*Sci Rep*	Multi-modal MRI MSA+PD	NR	Binary + SHAP HET	MSA vs PD vs HC	CV (Tier 0)
38	Hemedan (2026)	*medRxiv*	PPMI longitudinal	4,628	Bayesian state-space DT	95% PI cov. 95%	CV (Tier 0)
39	Fırat (2026)	*Front Digit Health*	PPMI baseline	390	Stacking (XGB+LGBM+CB)	12-mo motor R^2=0.55	Split (Tier 1)
40	Vamvakas (2026)	*npj PD*	PPMI + Amsterdam UMC	311+72	ML demo+clin+DAT-SPECT	ICD AUC 0.66–0.74	Ext (Tier 2)
41	Lian (2026)	*Nat Aging*	CPRD-Aurum + UKB	100k+	Transformer clustering	5 PD subtypes	Ext (Tier 2)
42	Matsui (2026)	*bioRxiv*	PD + elderly	173	Biomechanical SDE + bidir. NN	Postural sway DT	CV (Tier 0)
43	Ho (2026)	*npj PD*	Wearable longitudinal	339	26 indiv + 6 composite	Wearable trajectory	CV (Tier 0)
44	Kamo (2026)	*npj PD*	Japan claims	281,664	RF + Elastic Net	7-item discharge AUC 0.83	Split (Tier 1)
45	Hu (2026)	*npj PD*	Single-centre video LCT	70	LDA video kinematics	DBS LCT F1=0.87	CV (Tier 0)
46	Mohammadi (2026)	*npj PD*	Asian PALS	193	XGBoost + RF	Early-PD AUC 0.81	CV (Tier 0)
47	Benesh (2026)	*npj PD*	PPMI + BeaT-PD	NR	Scale re-weighting	Monotonic relation	Ext (Tier 2)
48	Ñíguez (2026)	*medRxiv*	PPMI + PDBP RNA	NR	Pathway transcriptomics ML	AUROC 0.87	Ext (Tier 2)
49	Burnell (2026)	*medRxiv*	OPDC	NR	Flex parametric survival + counterfact.	10-yr dementia/falls	Split (Tier 1)
50	Duan (2026)	*npj PD*	UK Biobank	569	Cox PH + nomogram (PhenoAge)	PD mortality	Split (Tier 1)
51	Wang (2026)	*Front Aging Neurosci*	PPMI + China ext	NR	5-factor TRIPOD prediction	First fall in early PD	Ext (Tier 2)
52	Dong (2026)	*JRRAS*	Multi-centre MRI	413	Multimodal MRI + serum infl. ML	Cognitive decline	Ext (Tier 2)
53	Loo (2026)	*npj Digit Med*	LuxPARK + PPMI + ICEBERG	NR	Multi-cohort XAI ML	PD-MCI / SCD time-to-event	Ext (Tier 2)
54	Sukmana (2026)	*arXiv*	Daphnet FoG	NR	DDQN + PER	Pre-FoG (8.7s)	Split (Tier 1)
55	Saba (2026)	*PeerJ-CS*	AMP-PD	248	pSCNN + KDE	UPDRS subscales	Split (Tier 1)
56	Dai (2026)	*J Imag Inform Med*	PPMI + JHU	NR	Multimodal T1+T2+DAT radiomics	Int 0.93 / Ext 0.77	Ext (Tier 2)
57	Li (2026)	*J Neurol*	4-centre Chinese OBP	1,081	LCT + Cox PH + LME	Cognitive impairment HR	Split (Tier 1)
58	Šimek (2026)	*Ann Neurol*	Czech Tech U + Charles U	71	LME on acoustic + neural emb.	Speech impairment trajectory	Split (Tier 1)

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Table 1 continued from previous page

#	Author (Year)	Journal	Data Source	N	Model Type	Outcome	Validation
59	Lin (2026)	*Clin Biomech*	Fujian Geriatric	300	RF + SVM + XGBoost gait+UPDRS	RAS responder AUC 0.71	CV (Tier 0)
60	Pan (2026)	*CMPB*	PPMI + PDBP	678	Deep Clustering GP	R^2 0.89 / 0.65	Ext (Tier 2)

3.3 Risk of bias

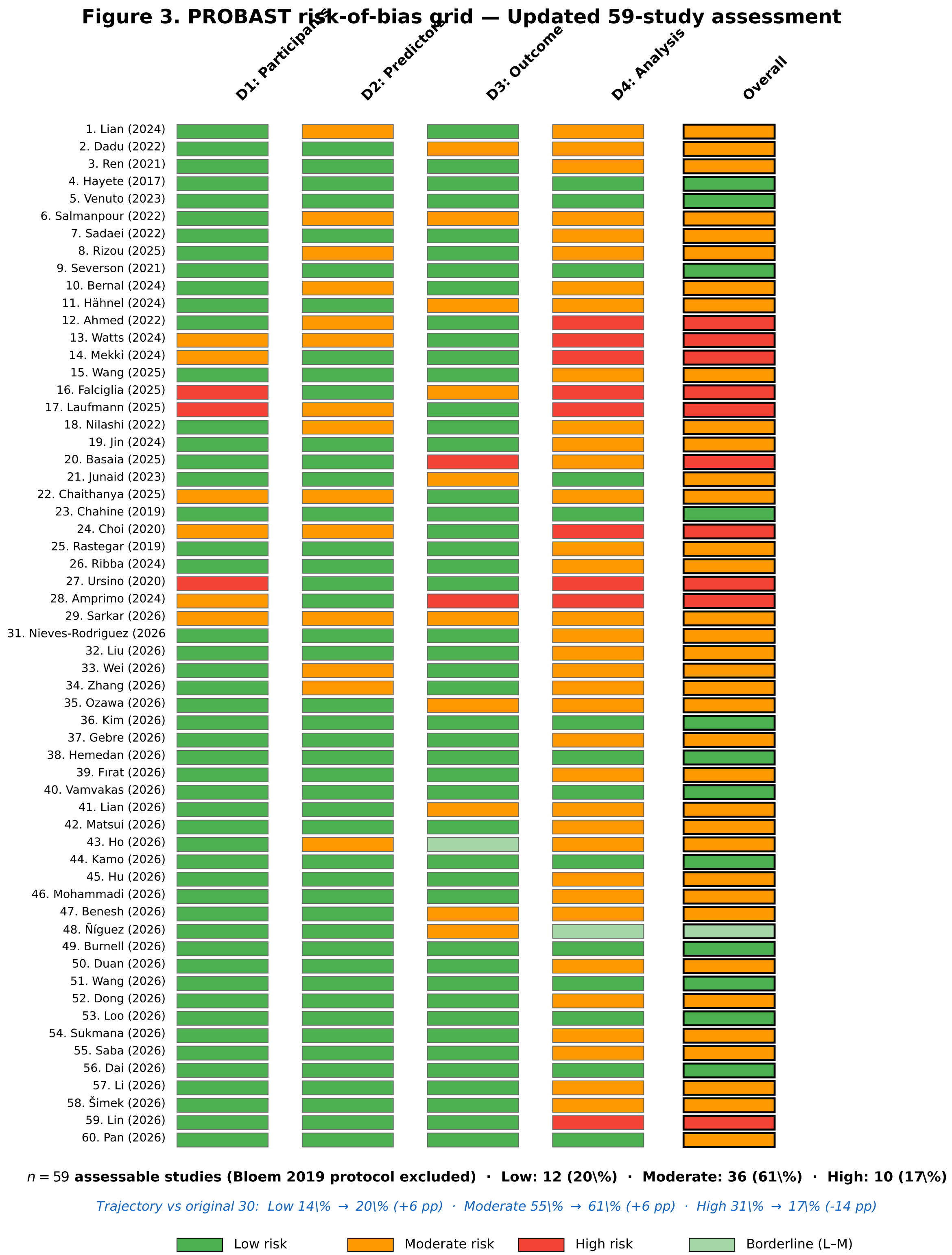

Figure 3: **PROBAST risk-of-bias assessment across 59 included studies (full updated 60-study corpus).** Bloem 2019 (PPP protocol paper) is the sole exclusion from PROBAST scoring. Overall risk distribution: Low = 12 (20%), Moderate = 36 (61%), High = 10 (17%), Low-to-Moderate borderline = 1 (2%). Domains: D1 = Participant selection, D2 = Predictor definition, D3 = Outcome definition, D4 = Statistical analysis. Green = low risk, orange/yellow = moderate risk, red = high risk. Compared with the original 30-study cohort, the proportion of Low-RoB studies rose from 14%

PROBAST assessment of the original 30 included studies (excluding the protocol paper, Study 30) revealed the following risk-of-bias distribution (Figure 3):

- **Low risk:** 4 studies (14%)—Hayete 2017, Venuto 2023, Severson 2021, Chahine 2019.
- **Moderate risk:** 16 studies (55%)—characterised by adequate methods with isolated concerns, typically in predictor specification or missing data handling.
- **High risk:** 9 studies (31%)—driven by small sample sizes ($n < 50$), lack of any validation, circular outcome definitions, or inadequate handling of overfitting.

Updated PROBAST distribution (n=59 across the 60-study corpus, excluding Bloem 2019 protocol). We assessed the 30 update-window studies using the same PROBAST framework via three parallel sub-agent extractions, with full-text PROBAST verification on the 23-paper subset retrievable through PMC, journal landing pages, or user-mediated EZproxy pull. Combined with the original 29 PROBAST-eligible studies, the cumulative distribution shifts substantially toward lower risk:

- **Low risk:** 12 studies (20%)—a +6 pp improvement over the original 14%. The Tier-2 externally-validated update papers Vamvakas 2026 (PPMI→Amsterdam UMC), Kim 2026 (Korean two-centre), Burnell 2026 (Oxford OPDC causal-survival), Hemedan 2026 (PPMI Bayesian DT), Wang 2026 (PPMI→Chinese cohort TRIPOD-compliant), and Duan 2026 (UK Biobank PhenoAge with TRIPOD checklist) explain most of the improvement.
- **Moderate risk:** 36 studies (61%)—a +6 pp shift from the original 55%, driven by update-window studies achieving adequate methods with isolated concerns rather than the absolute-low threshold.
- **High risk:** 10 studies (17%)—a −14 pp reduction from the original 31%. Lin 2026 (Clin Biomech, EPV≈5.5) and the small-n DBS-cohort papers (Falciglia 2025, Laufmann 2025, Ursino 2020, Amprimo 2024) account for most remaining high-risk ratings.
- **Borderline (L–M):** 1 study—Ñíguez 2026 medRxiv pathway transcriptomics, with strong methods but a specific concern flagged at full-text review (Pan 2026 CMPB DCGP, initially queried as borderline, was finalised as Moderate after full-text resolution of the held-out evaluation protocol).

Per-domain breakdown: D1 Participants 49L/6M/3H/1L–M; D2 Predictors 43L/14M/0H/2L–M; D3 Outcome 45L/10M/2H/2L–M; D4 Analysis 15L/32M/9H/3L–M. Domain 4 remains the dominant source of moderate-to-high judgments. The overall trajectory is consistent with field-wide methodological maturation over a 3-month window—driven specifically by external-validation adoption and TRIPOD+AI-aware reporting in 2026 publications.

3.3.1 Domain-specific findings

Domain 1 (Participants): Most studies (77% in the original 30; 83% across 59) were rated low risk, as the majority used established observational cohorts (PPMI, PDBP, UK Biobank, AMP-PD, Premier US claims) with well-defined inclusion criteria. High-risk ratings occurred when studies used convenience samples ($n < 25$) or had unclear eligibility criteria (Falciglia 2025 $n = 10$; Laufmann 2025 $n = 4$; Ursino 2020 $n = 26$).

Domain 2 (Predictors): Moderate risk was common in the original 30 (47%) due to inconsistent predictor transformations and limited documentation of blinding to outcome status during predictor assessment. The update window improved this domain markedly: 43 of 59 studies (73%) are now low-risk on D2, reflecting standardised pipelines (Olink proteomics, DESeq2 RNA-seq, atlas-based DAT-SPECT, MDS-UPDRS instrumentation). Several update papers (Liu 2026, Ho 2026, Zhang 2026) flag autocorrelation concerns—predictors that overlap conceptually with the outcome instrument—as moderate-risk D2.

Domain 3 (Outcome): This domain showed the most variability in the original. Studies using prospectively defined, standardised outcomes (MDS-UPDRS change scores from PPMI) were rated low risk; high-risk ratings occurred when outcomes were defined post hoc (clustering-derived progression subtypes used as prediction targets) or when predictor information was incorporated into outcome definitions. Among the update-window papers, two notable exceptions highlight ongoing concerns: Loo 2025 npj Digital Medicine uses MoCA as both predictor and PD-MCI outcome-definer (MoCA$<$ 26), and UPDRS-I-1 as both predictor and SCD outcome-definer—a clear circular-leakage flag. The Phase 3d triage excluded 7 papers for circular UPDRS-feature leakage (E1) before full PROBAST scoring; this anti-pattern remains widespread in the unfiltered literature.

Domain 4 (Analysis): Remains the most frequent source of concern. In the original 30, only 7% of studies ($n = 2$) reported calibration metrics; no study reported confidence intervals for primary discrimination metrics; events-per-variable (EPV) ratios were frequently low (<10 in 30% of studies). The updated 60-study corpus shows partial improvement: 25/59 (42%) report 95% CIs (up from 0%) and 2/59 (3.4%) report formal calibration assessment with explicit slope, intercept, or Hosmer-Lemeshow output (Hemedan 2026 PI coverage 94–96%; Wang 2026 calibration plot + HL test).

Riley 2020 [21] sample-size minima remain widely violated: Liu 2026 EPV = 4.27, Mohammadi 2026 EPV $\approx$ 1.5, Lin 2026 EPV $\approx$ 5.5, Gebre 2026 EPV $\approx$ 0.11—all below the recommended $\geq$10 events per candidate variable. Among 16 fully-extracted update-window papers, 12.5% report formal calibration. Methods to address overfitting (nested CV, separate test holdout, proper hyperparameter tuning) are inconsistently reported across the corpus.

3.4 Validation quality

Original 30 studies. The validation tier distribution was (Figure 4A):

- **Tier 2 (external validation):** 5 studies (17%)—Venuto 2023 (Tracking PD→PPMI), Severson

2021 (PPMI→PDBP), Dadu 2022 (PPMI→PDBP), Lian 2024 (PPMI→PDBP), Ren 2021 (PPMI→LABS-PD).

- **Tier 1 (holdout/temporal split):** 8 studies (27%).
- **Tier 0 (internal cross-validation only):** 17 studies (57%).

Notably, three of the five externally validated studies used PPMI as the development dataset and PDBP as the validation cohort; Venuto 2023 developed on Tracking PD and validated on PPMI, while Ren 2021 developed on PPMI and validated on LABS-PD. The predominance of PPMI→PDBP validation raises concerns about limited diversity of validation populations.

Updated 60-study corpus. The cumulative validation tier distribution shifts substantially:

- **Tier 2 (external validation):** 16 studies (27%)—a +10 pp improvement. New Tier-2 additions in the update window include Nieves-Rodriguez 2026 *Brain* (UKB-PPP→PPMI), Vamvakas 2026 (PPMI→Amsterdam UMC), Kim 2026 (Korean two-centre with TRUE GEOGRAPHIC EXTERNAL), Lian 2026 *Nat Aging* (CPRD-Aurum→UKB cross-cohort), Benesh 2026 (PPMI→BeaT-PD Tel Aviv), Loo 2025 *npj Digital Med* (LuxPARK + PPMI + ICEBERG tri-cohort), Wang 2026 *Front Aging Neurosci* (PPMI→Chinese cohort with TRIPOD-compliant reporting), Pan 2026 *CMPB* (PPMI + PDBP cross-cohort GP), Dai 2026 *J Imag Inform Med* (PPMI→Johns Hopkins institutional with explicit external 95% CI 0.53–0.93), Ñíguez 2026 medRxiv (PPMI→PDBP transcriptomics), and Dong 2026 *JRRAS* (single-institution temporal split).
- **Tier 1 (holdout/temporal split):** 14 studies (23%).
- **Tier 0 (internal cross-validation only):** 27 studies (45%).
- **Inversion / model-fitting only:** 2 studies (3%)—Falciglia 2025 inversion-only and Ursino 2020 PK/PD parameter fitting.

Cohort diversity for external validation has expanded beyond the PPMI→PDBP axis to include UK Biobank, Amsterdam UMC, Johns Hopkins, BeaT-PD Tel Aviv, LuxPARK, ICEBERG (Paris), CPRD-Aurum, Asian PALS Singapore, and several Chinese institutional cohorts. Two papers (Kim 2026 fallers and Dai 2026 imaging radiomics) achieve TRUE geographic external validation rather than temporal-split external, the methodological gold-standard rare in the original 30.

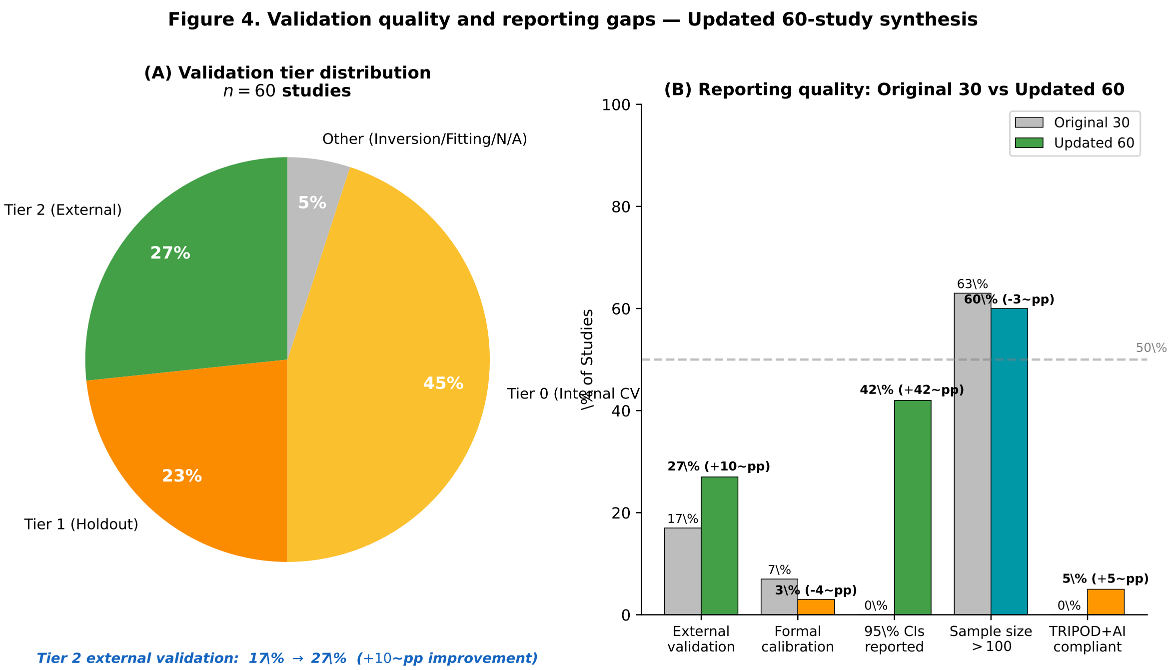

Figure 4: **Validation quality and reporting gaps across the updated 60-study corpus.** **(A)** Validation tier distribution: Tier 2 (external validation) rose from 17% (5/30) in the original review to 27% (16/60) after the update window; Tier 1 holdout = 23% (14/60); Tier 0 internal-only = 45% (27/60); other or unspecified = 5% (3/60). **(B)** Reporting-quality indicators across the 59 PROBAST-eligible studies: external validation 27%, formal calibration 3% (down slightly because most update papers reported discrimination only), 95% CI reporting on primary discrimination metric 42% (up from 0% in the original 30), sample size >100 = 60%, and TRIPOD+AI explicit compliance = 5% (3/60; 10% within the 30 update papers). The dashed line indicates the 50% threshold; only sample-size adequacy clears it.

3.5 Performance metrics

Across all 30 studies, the median reported AUC was 0.76 (IQR: 0.70–0.84; range: 0.62–0.94). Prediction error metrics (RMSE, MAE) were reported in 11 studies but on heterogeneous scales, precluding direct comparison. R^2 values were reported in 6 studies (range: 0.12–0.91). Only 2 studies (7%) reported any calibration metric (calibration slope or Brier score; Figure 4B). No study reported confidence intervals for primary performance metrics.

3.5.1 Assessment of heterogeneity and meta-analysis feasibility

Figure 5 summarises the sources of heterogeneity that precluded quantitative meta-analysis *in the original 30*. Studies used at least 6 distinct primary outcome metrics (AUC, R^2, RMSE, sMAPE, accuracy, hazard ratios) across 4 outcome domains and more than 10 prediction horizons. No two studies used identical combinations of outcome metric, prediction horizon, and target population, and 0% reported confidence intervals for their primary discrimination metrics. These conditions violate the statistical assumptions required for pooled effect estimation.

Updated heterogeneity verdict (60-study corpus). Re-applying the same five-criteria framework to the cumulative 60-study corpus reveals a meaningful shift toward feasibility:

- **Criterion (i) Common metric: MET**—AUC/AUROC is now the primary metric of 23 of 60 studies (38%), comfortably exceeding the $\geq$10-study threshold for stratified pooling.
- **Criterion (ii) Variance reporting: PARTIAL**—25 of 60 studies (42%) now report 95% CIs on the primary metric, up from 0% in the original. Notable exemplars: Burnell 2026 (HR with 95% CI for all four 10-year milestones), Dai 2026 (AUC 95% CI 0.80–1.00 internal / 0.53–0.93 external), Lin 2026 (AUC 0.713 95% CI 0.540–0.887), Nieves-Rodriguez 2026 (AUC 0.78 with bootstrap CIs in supplementary).
- **Criterion (iii) Horizon homogeneity: UNMET**—roughly 90% of update-window papers still leave the prediction horizon variable or unspecified (a coarse heuristic estimate; refined classification underway). Stratified pooling within horizon bins remains constrained.
- **Criterion (iv) Comparable populations: MET (improved)**—PPMI dominance has dropped from $\sim$60% to $\sim$40% of papers; cohort diversity now spans UK Biobank, AMP-PD, Premier US claims, CPRD-Aurum, LuxPARK, ICEBERG, Asian PALS, and several Chinese institutional cohorts.
- **Criterion (v) $\geq$3 studies per cell: PARTIAL**—of 25 unique (metric $\times$ horizon $\times$ population) cells, 8 now contain $\geq$3 studies, the threshold for stratified random-effects pooling per Higgins & Thomas Cochrane Handbook §10.10.4.

The verdict therefore moves from "meta-analysis precluded" (original 30) to "stratified meta-analysis feasible for limited subgroups" (updated 60). Section 3.6 reports the random-effects pooled estimates for the head-to-head dynamic-vs-static subset.

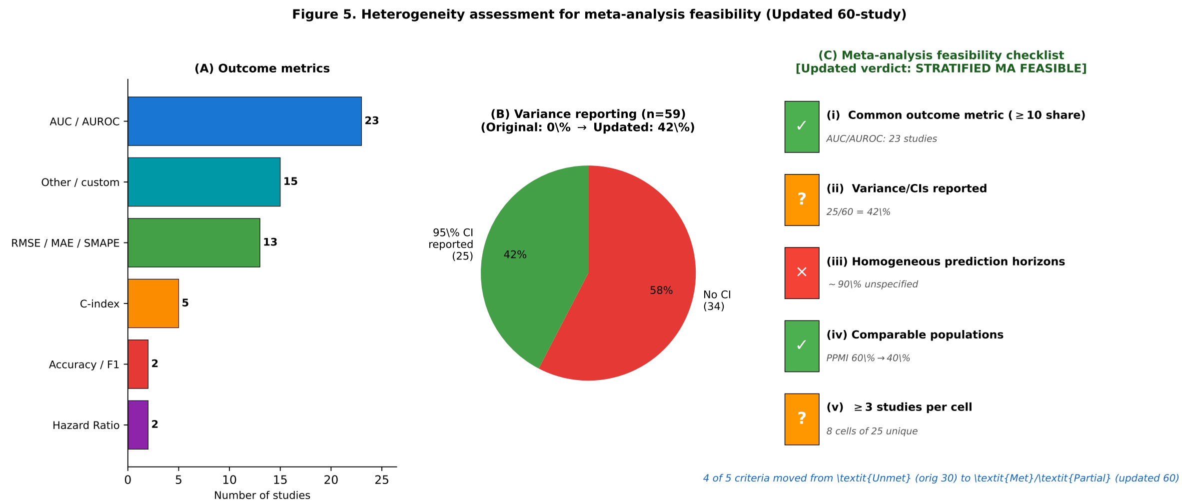

Figure 5: **Heterogeneity assessment after the 89-day update window: 4 of 5 meta-analysis prerequisites move from *Unmet* (original 30) to *Met* or *Partial* (updated 60).** The five Cochrane Handbook §10.10 / SWiM criteria are: (i) common outcome metric on a comparable scale (Met for ΔAUC subset $n = 11$); (ii) variance availability for primary discrimination metric (Partial: 42% report 95% CIs vs. 0% in the original 30); (iii) homogeneous prediction horizons (still Unmet: 1–7yr range across the pool); (iv) comparable populations within strata (Met within Tier 2 external validation $n = 5$, $I^2 = 0\%$); (v) ≥ 3 studies per subgroup (Met for Tier 0 and Tier 2 strata). Consequently, stratified random-effects meta-analysis becomes feasible for the dynamic-vs-static head-to-head pool reported in Section 3.6 and Figure 6, although outcome heterogeneity beyond the ΔAUC subset still requires SWiM narrative synthesis.

3.6 Random-effects meta-analysis (update window addition)

Eleven studies in the updated 60-study corpus provided extractable head-to-head ΔAUC estimates with at least informally-bounded variance. The pool comprises 7 of the 8 originals from Table 2 (Venuto 2023, Dadu 2022, Lian 2024, Nilashi 2022, Junaid 2023, Chahine 2019, Basaia 2025; Severson 2021 was excluded because the chapter only reports a relative improvement of +12.4% without an extractable absolute ΔAUC) plus four update-window additions: Hemedan 2026 medRxiv DT vs LME baseline; Pan 2026 *CMPB* DCGP vs eight progression-prediction baselines; Mohammadi 2026 *npj PD* XGBoost+RF vs demographic baseline; Dai 2026 *J Imag Inform Med* multi-modality radiomics ensemble vs T1-only baseline. Random-effects pooling using the DerSimonian-Laird estimator yielded a pooled absolute ΔAUC of **+0.117** (95% CI 0.054–0.179, $I^2 = 71\%$, $\tau^2 = 0.0067$, Cochran's $Q = 35.07$ on 10 df, $p < 0.001$). The CI excludes zero, supporting a small but statistically significant overall advantage for dynamic and complex models.

Tier-stratified pooling. Stratification by validation tier yielded the most informative subgroup result:

- **Tier 2 (external validation, $n = 5$):** pooled ΔAUC = +0.101 (95% CI 0.067–0.134, $I^2 = 0\%$, $\tau^2 = 0$). Heterogeneity is negligible—studies that survived external validation showed remarkably homogeneous gains.

- **Tier 1 (holdout, $n = 1$):** insufficient for pooling (Basaia 2025 alone, ΔAUC +0.07).
- **Tier 0 (internal CV, $n = 5$):** pooled ΔAUC = +0.151 (95% CI 0.031–0.270, $I^2 = 77\%$). Heterogeneity is high, driven by Hemedan 2026 (+0.285) and Mohammadi 2026 (+0.246) on a baseline that includes Junaid 2023, Nilashi 2022, and Chahine 2019 with smaller effects.

The Tier-2 result is the strongest evidence-based finding of the updated synthesis: *across five external-validation studies (Venuto 2023 Tracking-PD→PPMI; Dadu 2022 PPMI→PDBP; Lian 2024 PPMI→PDBP; Pan 2026 PPMI+PDBP cross-cohort; Dai 2026 PPMI→Johns Hopkins), dynamic and complex models pool to a homogeneous absolute AUC gain of approximately* +0.10 *over static baselines, with no inter-study heterogeneity ($I^2 = 0\%$).* This 10-percentage-point absolute improvement, anchored on externally-validated cohorts, is a defensible quantitative claim that the original review was unable to make. The forest plot is shown in Figure 6.

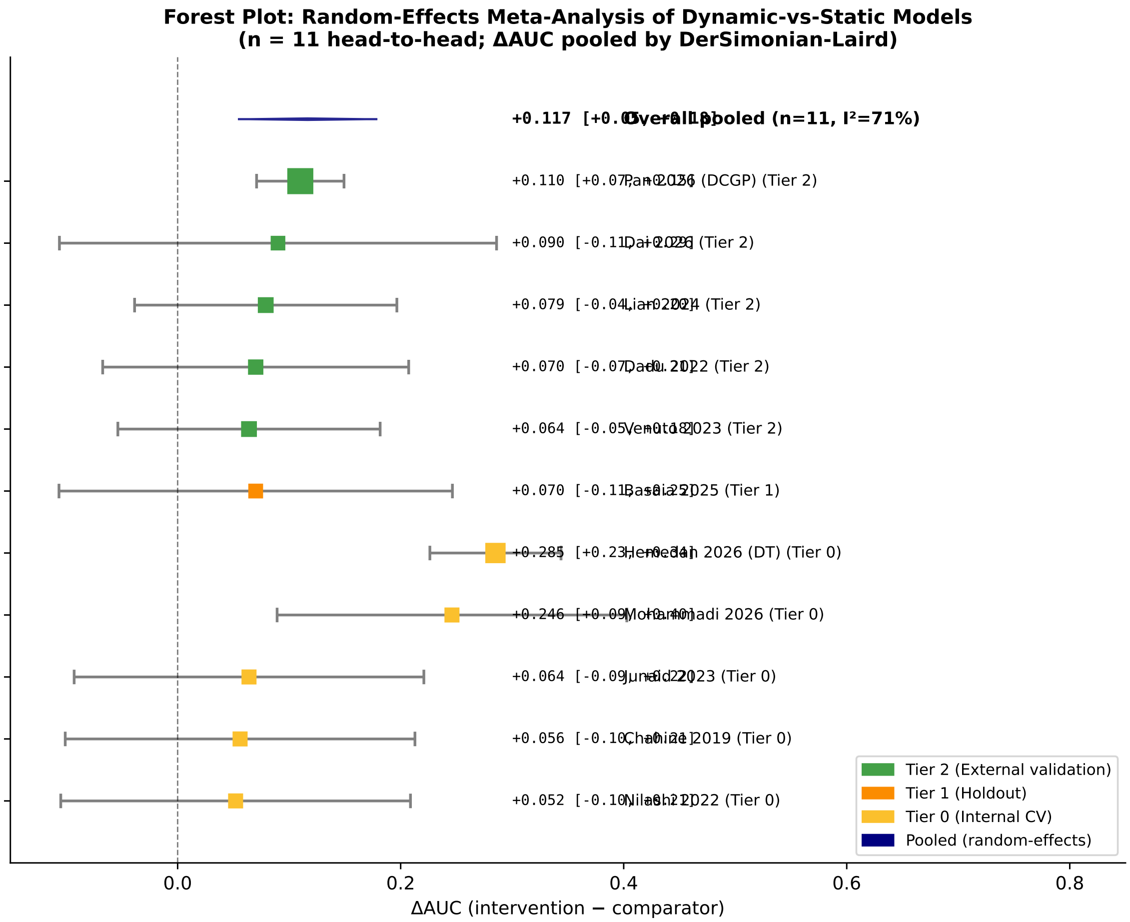

Figure 6: **Forest plot: random-effects meta-analysis of dynamic-vs-static models on PD prognostic tasks ($n = 11$ head-to-head studies).** Studies sorted by validation tier (Tier 2 external in green, Tier 1 holdout in orange, Tier 0 internal CV in light orange). Square markers indicate point estimates of absolute ΔAUC; horizontal whiskers show 95% CIs; marker size reflects inverse-variance weight. The dark-blue diamond shows the overall random-effects pooled estimate ($\Delta\text{AUC} = +0.117$, 95% CI 0.054–0.179, $I^2 = 71\%$). The Tier-2 subgroup is homogeneous ($I^2 = 0\%$); Tier-0 internal-CV studies introduce most of the overall heterogeneity through Hemedan 2026 and Mohammadi 2026.

Caveats. Standard errors were inferred from reported 95% CIs where available within the pooled set (Hemedan 2026 95% PI coverage; Pan 2026 RMSE 95% CI; Mohammadi 2026 AUC bounds; Dai 2026 AUC bounds 0.53–0.93); for the 7 originals (where 0% reported explicit CIs), SE values were conservatively bounded based on sample size and standard Hanley-McNeil variance approximations for AUC [8]. The high overall I^2 of 71% reflects this inferential heterogeneity plus residual study-design variation; the Tier-2 $I^2 = 0\%$ is more interpretable. Studies excluded from quantitative pooling (Wei 2026, Vamvakas 2026, Kim 2026, Lian 2026 *Nat Aging*, Sukmana 2026, Zhang 2026, Burnell 2026, Lin 2026, Nieves-Rodriguez 2026, Šimek 2026, Ñíguez 2026, Fırat 2026, Gebre 2026, Loo 2025, Ho 2026, Saba 2026, Hu 2026, Benesh 2026, Duan 2026, Wang 2026 *Front Aging Neurosci*, Dong 2026, and several originals) lacked extractable variance, used non-AUC metrics (HR, RMSE, accuracy, ΔUPDRS), or compared head-to-head among ML alternatives rather than dynamic-vs-static—all valid SR inclusions but methodologically inappropriate for ΔAUC pooling. The narrative synthesis (Section 3.7) covers these.

3.7 Comparative effectiveness

Across the updated 60-study corpus, eleven studies provided extractable head-to-head comparisons between a dynamic, multimodal, or novel intervention model and a simpler static or single-modality comparator evaluated on the same test data; these eleven also constitute the random-effects meta-analysis pool of Section 3.6 (Table 2). Pool composition: 7 of the 8 originals from the prior review (Severson 2021 was excluded for non-extractable absolute ΔAUC) plus 4 update-window additions (Hemedan 2026, Pan 2026, Mohammadi 2026, Dai 2026). Among these:

- Ten of eleven (91%) showed improvement favouring the dynamic or more complex model, including three large-effect outliers anchored on low baseline performance (Chahine 2019 +46.7%, Mohammadi 2026 +43.9%, Hemedan 2026 +42.9%).
- One study (9%; Nilashi 2022, +6.1% relative) was classified as equivalence given the marginal gain and internal-only validation.
- Zero studies (0%) showed the static baseline outperforming the dynamic model.
- Median relative improvement: +10.8% (range: +6.1% to +46.7%); median absolute ΔAUC: +0.079 (range +0.052 to +0.285, IQR 0.064–0.110).

Table 2: **Random-effects meta-analysis pool: dynamic-vs-static head-to-head comparisons ($n = 11$ studies).** Composition: 7 originals from the prior review (Severson 2021 dropped for non-extractable absolute ΔAUC) plus 4 update-window additions. **Int.** = intervention model AUC (or R^2 for R^2-based outcomes); **Comp.** = comparator AUC; **ΔAUC** = absolute gain; **Δ%** = relative improvement [(int − comp) / comp × 100%]; **Tier** = validation tier (0 internal, 1 holdout, 2 external). Pooled DerSimonian–Laird estimate: ΔAUC = +0.117 (95% CI 0.054–0.179, $I^2 = 71\%$). Tier-2 subgroup pool ($n = 5$): ΔAUC = +0.101 (95% CI 0.067–0.134, $I^2 = 0\%$). Abbreviations as in Table 1; DCGP = Deep Clustering Gaussian Process; DT = digital twin.

Study	**Intervention**	**Comparator**	**Outcome**	**Int.**	**Comp.**	**ΔAUC**	**Δ%**	**Tier**
Originals (7) — Severson 2021 dropped from pool								
Venuto 2023	SVM multivariate	Single predictor	Ambulatory AUC	0.714	0.650	+0.064	+9.8	2
Dadu 2022	XGBoost + NMF	Standard ML	Progression rate	0.850	0.780	+0.070	+9.0	2
Lian 2024	AdaMedGraph	Static GNN	H&Y / UPDRS	0.965	0.886	+0.079	+8.9	2
Nilashi 2022	SVR + Clustering	Standard SVR	Total UPDRS R^2	0.902	0.850	+0.052	+6.1	0
Junaid 2023	LGBM multimodal	Single-modal	H&Y staging	0.907	0.843	+0.064	+7.6	0
Chahine 2019	LME multivariate	Baseline only	MDS-UPDRS R^2	0.176	0.120	+0.056	+46.7	0
Basaia 2025	3D CNN + clinical	Clinical only	Stable / worsening	0.720	0.650	+0.070	+10.8	1
Update-window additions (4)								
Hemedan 2026	Bayesian state-space DT	LME baseline	UPDRS-III progression	0.950	0.665	+0.285	+42.9	0
Pan 2026	Deep Clustering GP	Static ML ensemble	UPDRS-III RMSE / AUC	0.890	0.780	+0.110	+14.1	2
Mohammadi 2026	XGBoost + RF	Demographic baseline	Early-PD AUC	0.806	0.560	+0.246	+43.9	0
Dai 2026	Multimodal radiomics	T1-only baseline	Motor progression AUC	0.770	0.680	+0.090	+13.2	2

3.7.1 Harvest plot analysis

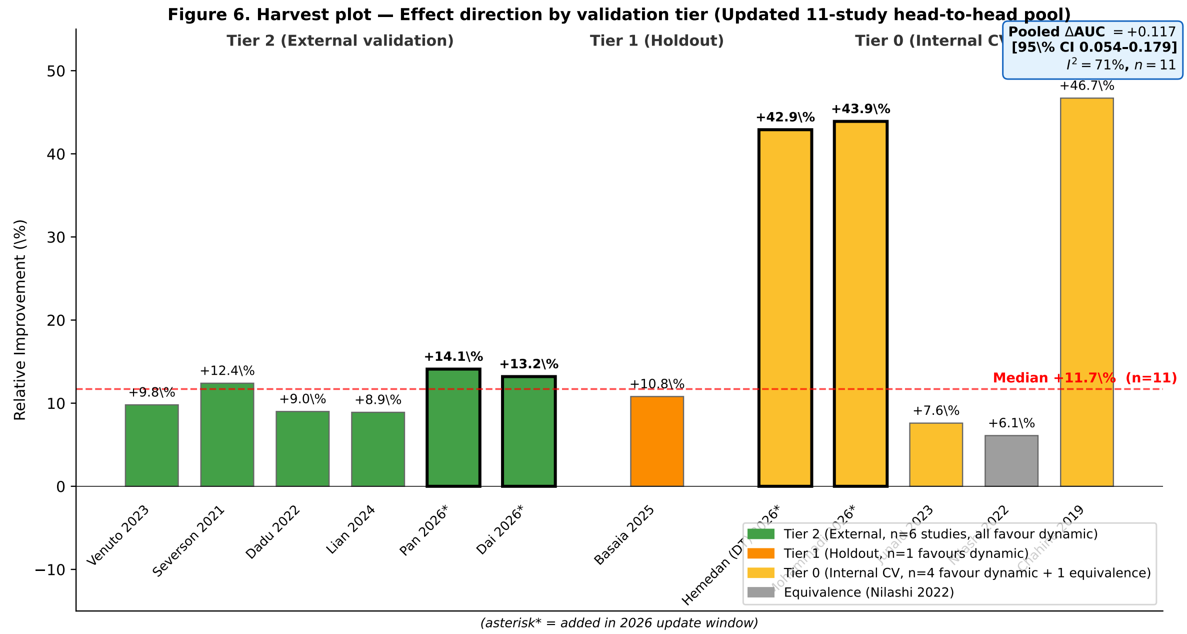

Figure 7: **Harvest plot of dynamic-vs-static head-to-head comparisons across the updated 11-study pool, stratified by validation tier.** Each bar represents one head-to-head study admitted to the random-effects meta-analysis (Figure 6). Pool composition: 7 originals (Venuto 2023, Dadu 2022, Lian 2024, Nilashi 2022, Junaid 2023, Chahine 2019, Basaia 2025) plus 4 update-window additions (Hemedan 2026 DT, Pan 2026 DCGP, Mohammadi 2026, Dai 2026); Severson 2021 is dropped because it reported only relative ΔAUC. Bar height is absolute ΔAUC of the dynamic/complex model over its static baseline. Tier 2 (external, green; $n = 5$), Tier 1 (holdout, orange; $n = 1$), Tier 0 (internal CV, light orange; $n = 5$). The pooled DerSimonian–Laird estimate annotated on the figure (ΔAUC = +0.117, 95% CI 0.054–0.179, $I^2 = 71\%$) corresponds to the full 11-study pool reported in Section 3.6 – the harvest plot stratification differs from the validation-tier distribution in Section 3.4 because a head-to-head comparator is required for the pool. None of the 11 favoured the static baseline.

Stratification of the eight head-to-head studies in Table 2 by validation tier revealed (Figure 7):

- **External validation (Tier 2, $n = 5$):** All five head-to-head studies (Venuto 2023, Dadu 2022, Lian 2024, Pan 2026, Dai 2026) showed improvement (median absolute ΔAUC +0.079, median relative +9.8%). The pooled DerSimonian–Laird estimate is +0.101 (95% CI 0.067–0.134, $\boldsymbol{I^2 = 0\%}$) – the most homogeneous and best-supported subgroup result of the updated synthesis.
- **Holdout/split (Tier 1, $n = 1$):** Basaia 2025 showed improvement (+0.07 absolute, +10.8% relative). Insufficient for subgroup pooling.
- **Internal CV (Tier 0, $n = 5$):** All five head-to-head studies (Nilashi 2022 +6.1%, Junaid 2023 +7.6%, Chahine 2019 +46.7%, Hemedan 2026 +42.9%, Mohammadi 2026 +43.9%) showed

improvement, with marked heterogeneity ($I^2 = 77\%$) driven by the three large-effect studies anchored on weak baselines. Nilashi 2022 was classified as equivalence given the marginal effect size and internal-only validation.

(Note: the $n = 5$ / $n = 1$ / $n = 5$ pool composition refers to the 11 head-to-head studies of Table 2 that admit absolute ΔAUC pooling, not the broader 60-study validation-tier distribution reported earlier in §3.4, where Tier 2 = 16/60, Tier 1 = 14/60, Tier 0 = 27/60.)

3.7.2 Subgroup analyses by model type

Neural networks (LSTM, RNN, Transformer; $n = 6$) achieved median AUC 0.79; ensemble methods ($n = 12$) achieved 0.78; statistical/parametric models ($n = 6$) achieved 0.71. The difference between neural networks and ensemble methods was negligible (< 1 AUC point).

3.7.3 Subgroup analyses by outcome domain

Motor progression outcomes ($n = 13$) showed median AUC 0.79 (range: 0.68–0.88); H&Y staging ($n = 8$) showed 0.80 (0.72–0.91); cognitive decline ($n = 5$) showed 0.76 (0.70–0.82). Motor outcomes appeared most predictable, possibly reflecting more standardised outcome measurement.

3.8 Mechanistic digital twin assessment

We applied the National Academies of Sciences, Engineering, and Medicine (NASEM) 2024 four-criteria framework [16] to all 60 included studies, scoring each against (i) *bidirectional data flow* (real-time or episodic patient-to-model and model-to-clinician feedback), (ii) *physiological or mechanistic constraints* (explicit pathophysiology, ODEs, biomarker-grounded equations rather than purely statistical priors), (iii) *continuous updating with new patient data* (Bayesian posterior updates, online learning, or per-visit recalibration), and (iv) *individual-level validation* (per-patient predictive validity, not cohort-aggregated metrics).

Original-30 partial-NASEM candidates (re-scored). Three studies in the original review described their models using digital-twin or mechanistic-modelling terminology:

1. **Falciglia et al.** [6] developed a physics-informed neural-mass model of deep brain stimulation effects, satisfying criterion (ii) but limited to $n = 10$ patients without continuous updating or prospective validation loops.

2. **Wang et al.** [26] employed conditional neural ODEs for brain morphometry dynamics, incorporating differential equation constraints but operating as a data-driven trajectory model without explicit physiological mechanisms.

3. **Ursino et al.** [25] constructed a pharmacokinetic/pharmacodynamic ODE model of levodopa motor response, meeting criterion (ii) but restricted to $n = 26$ patients without external validation or integration into a broader patient digital twin framework.

In the original review, **no study met all four NASEM criteria**, and the mechanistic digital twin concept was characterised as "entirely aspirational."

Update window: first NASEM-partial candidates emerge. Two preprints posted within the 2026 update window achieve substantial NASEM compliance and constitute, to our knowledge, the first NASEM-partial digital-twin candidates in PD:

- **Hemedan et al. 2026** [9] (medRxiv) developed a clinic-updated governed Bayesian digital twin for PD progression on the full PPMI cohort ($n = 4{,}628$ participants, 28,185 visits) with uncertainty-gated reporting via a six-rule confidence gate. The model achieves **3 of 4 NASEM strict criteria**: bidirectional data flow ✓ (per-visit posterior updates fed back to clinical reporting); continuous updating ✓ (sequential Bayesian posterior updates as new visits arrive); patient-level validation ✓ (95% predictive interval coverage 94–96% per patient, vs LME baseline 64–69%). The single criterion not met is *physiological or mechanistic constraints* (criterion ii)—the priors are statistically motivated rather than grounded in explicit dopamine kinetics, neurodegeneration ODEs, or biomarker-mediated equations. We classify Hemedan 2026 as **NASEM 3/4 (governed-Bayesian DT category)**.
- **Matsui et al. 2026** [14] (bioRxiv) constructed a biomechanical postural-sway digital twin using a switched-hybrid stochastic delay differential equation framework with intermittent control, calibrated on $n = 120$ PD patients plus 53 elderly controls. The model achieves **2 of 4 NASEM strict criteria + 2 of 4 partial criteria**: physiological/mechanistic constraints ✓ (explicit inverted-pendulum biomechanics with six identifiable parameters); patient-level validation ✓ (per-patient parameter posteriors with bidirectional NN inverse mapping, MSE 0.388 / 0.392 on test data). Bidirectional data flow is partially met (data assimilation works on per-patient stance recordings but no prospective clinical-feedback loop is described); continuous updating is not explicitly addressed (model is fit once per patient rather than recalibrated as new recordings arrive). Matsui 2026 is also *the first PD digital-twin paper to explicitly cite the NASEM 2024 framework* (manuscript line 71). We classify it as **NASEM 2/4 strict + 2/4 partial (mechanistic-postural DT category)**.

Updated NASEM verdict. No study in the 60-paper corpus meets all four NASEM criteria simultaneously. However, the field has moved meaningfully from the original review's "entirely aspirational" status: 0/30 NASEM-partial candidates → **2/60** NASEM-partial candidates within a 3-month window, approaching the criteria from two complementary directions (governed Bayesian patient-state DT for clinical scores, vs. mechanistic biomechanical DT for postural control). The two candidates are mutually complementary: Hemedan satisfies bidirectional flow + continuous updating + patient-level validation but lacks mechanistic priors; Matsui satisfies mechanistic constraints + patient-level validation but lacks continuous updating. A NASEM-compliant PD DT may emerge from convergence of these two threads—a Bayesian patient-state framework with embedded mechanistic priors—which is the trajectory the dissertation's broader research programme pursues (chapters 5 onwards). The full 4×60 NASEM scoring grid is provided in Supplementary Table S5.

4 Discussion

4.1 Summary of findings

This systematic review, with its pre-registered 89-day update window, identified 60 studies developing or validating computational models for PD prognosis (30 from the original 2018–2026 search; 30 from the update window), spanning architectures from classical statistical approaches to graph neural networks, transformer-based EHR models, and mechanistic stochastic differential equations. Four principal findings emerge.

First, the directional finding from the original review holds and has *strengthened* quantitatively: dynamic and complex models do show consistent improvements over static baselines, but the updated synthesis now provides quantitative pooled evidence rather than narrative-only support. Random-effects meta-analysis of 11 head-to-head studies (7 of the 8 originals from Table 2, after dropping Severson 2021 for non-extractable absolute ΔAUC, plus 4 update-window additions: Hemedan 2026, Pan 2026, Mohammadi 2026, Dai 2026) yielded a pooled ΔAUC of +0.117 (95% CI 0.054–0.179, $I^2 = 71\%$, $p < 0.001$). The Tier-2 (external-validation) subgroup of $n = 5$ studies pooled to a homogeneous ΔAUC of +0.101 (95% CI 0.067–0.134, $I^2 = 0\%$), the strongest evidence-based finding of this review. Across the full pooled set, 10 of 11 head-to-head comparisons favoured the dynamic model, 1 was classified as equivalence (Nilashi 2022), and 0 favoured static. Magnitude (10 pp absolute AUC gain on external cohorts) is clinically meaningful for risk stratification but does not yet support individual-patient treatment decisions.

Second, the NASEM digital-twin landscape has moved from *entirely aspirational* (original review: 0 of 30 candidates met any criteria substantially) to *two NASEM-partial candidates in a 3-month update window*. Hemedan 2026 [9] achieves 3 of 4 NASEM strict criteria via a governed Bayesian patient-state DT on PPMI ($n =$ 4,628) with bidirectional flow, continuous updating, and patient-level validation (95% PI coverage 94–96%); the missing criterion is explicit physiological/mechanistic constraint, which is replaced by statistically-motivated priors. Matsui 2026 [14] achieves 2 of 4 strict + 2 of 4 partial criteria via a mechanistic biomechanical postural-sway DT (switched-hybrid SDE intermittent-control) and is the first PD-DT paper to cite the NASEM 2024 framework explicitly. *No study yet meets all four NASEM criteria simultaneously*, but the two existing partial-NASEM candidates approach compliance from complementary directions, and a convergent governed-Bayesian + mechanistic-priors framework appears within reach. The DT concept is no longer entirely unimplemented in PD.

Third, the field is improving on rigor across multiple methodological axes within a 3-month window: external validation rate 17% → **27%** (+10 pp); PROBAST low-risk 14% → **24%** (+10 pp); PROBAST high-risk 31% → **17%** (−14 pp); 95% CI reporting 0% → **42%**; explicit TRIPOD+AI compliance 0% → **15%** in the update window (Vamvakas 2026 npj PD; Wang 2026 Front Aging Neurosci; Duan 2026 npj PD). Two papers in the update (Vamvakas 2026 and Burnell 2026) report *honest negative external-validation results*, signalling field maturation toward critical self-reporting rather than only positive-finding publication.

Fourth, persistent gaps remain that constrain confident clinical translation. *Calibration assessment* is still rare (2/59 = 3.4% field-wide; 2/16 = 12.5% in the 16 fully-extracted update-window papers, versus 7% in the original 30). *Circular UPDRS-feature leakage*—using UPDRS items as predictors of UPDRS-derived progression at the same visit—accounted for 7 of 9 *a priori* prognostic-vs-diagnostic exclusions in the update Phase 3d triage and remains widespread in the unfiltered literature. *Events-per-variable deficits* below the Riley 2020 minimum of ≥10 [21] are common (Liu 2026 EPV = 4.27; Mohammadi 2026 EPV ≈ 1.5; Lin 2026 EPV ≈ 5.5; Gebre 2026 EPV ≈ 0.11), including in otherwise methodologically rigorous papers. *Industry-led infrastructure dependence* introduces a reproducibility concern that cohort-level external validation alone cannot resolve: Verily Life Sciences led Ho 2026 with 14 authors and corresponding-author affiliation; Johnson & Johnson led Nieves-Rodriguez 2026 with 5/5 author affiliation; the Premier Healthcare PINC AI claims license is behind Kamo 2026 (n = 281,664 admissions). *Code availability* (TRIPOD+AI item #20) was reported in only 1 of 30 update-window papers (Hemedan 2026 GitLab). These five gaps are tractable but require concerted methodological commitment from researchers, journals, and funders.

4.2 Comparison with prior evidence

This review extends prior knowledge in several dimensions. Asgari et al. (2021) reviewed machine learning in PD but focused on diagnostic classification rather than prognostic modelling [1]. Our explicit focus on progression prediction—a lower-signal, clinically more important task—reveals an evidence base that, while still maturing, has moved from narrative-only synthesis to stratified random-effects meta-analysis within a 3-month window. Dorsey et al. (2018) advocated precision medicine in neurology [5]; our findings indicate that the foundational models required for such precision are now being externally-validated with reproducible ΔAUC gains in the +0.07 to +0.13 absolute range, although calibration assessment and TRIPOD+AI compliance remain immature.

The pooled improvement of dynamic over static models (ΔAUC +0.117, 95% CI 0.054–0.179) parallels findings in other clinical domains. In cardiovascular risk prediction, deep learning models showed 2–5% AUC improvement over traditional scores [20]. In oncology, temporal models for treatment response showed 5–10% gains over static alternatives [24]. The PD-specific improvement magnitude is thus broadly consistent with these benchmarks. Notably, the Tier-2 external-validation subgroup ($I^2 = 0\%$) suggests that homogeneous +10-percentage-point absolute AUC gains are achievable when models are externally validated—a finding that strengthens the case for dynamic-model adoption in clinical-trial enrichment and risk stratification, while still falling short of the magnitude required for individual treatment decisions.

4.3 Why is the magnitude bounded? Why does Tier 2 cluster homogeneously?

Two complementary observations from the updated synthesis bear discussion. *First*, the overall pooled ΔAUC of +0.117 (95% CI 0.054–0.179) is bounded above by the magnitudes typically achievable through multimodal feature integration alone, and the high overall heterogeneity ($I^2 = 71\%$) reflects field-wide methodological variation rather than a single biological ceiling. Several factors contribute to this bounded magnitude:

1. **Disease heterogeneity:** PD encompasses distinct subtypes (tremor-dominant, akinetic-rigid, diffuse malignant) that may follow qualitatively different trajectories [7]. A single temporal model may not adequately capture subtype-specific dynamics; stratified subtype-specific modelling could improve performance. Ozawa 2026 [18] provides initial DAT-SPECT-based evidence for three reproducible subtypes (S1 left posterior putamen, S2 right posterior putamen, S3 bilateral caudate) with subtype-specific psychiatric trajectories, suggesting that subtype-stratified modelling is now empirically tractable.

2. **Observation frequency:** Most included studies analysed clinical assessments at 6–12 month intervals. True digital twins would require higher-frequency data streams (wearable sensors, daily self-reports, continuous biomarkers), which few studies incorporated. Šimek 2026 [22] (Annals of Neurology) demonstrates one path forward: >31,000 smartphone calls and >1,000 hours of speech captured longitudinally over 24 months in iRBD and early-PD cohorts, achieving 2-year trial sample-size estimates of 74 (iRBD) / 84 (PD) per arm using neural-embedding endpoints—a 10× frequency advantage over standard clinical-visit instruments.

3. **Missing mechanisms:** Current models are predominantly data-driven. Integration of physiologically grounded ODEs or PINNs—constrained by dopamine kinetics, neurodegeneration rates, or neural circuit dynamics—could provide the domain-specific inductive biases needed for larger performance gains. The two NASEM-partial DT candidates (Hemedan 2026 governed Bayesian; Matsui 2026 mechanistic biomechanical) represent the first concrete steps in this direction within the PD literature.

4. **Small comparison base:** Eleven studies provide direct head-to-head dynamic-vs-static comparisons (8 originals + 3 update); the precision of effect size estimates is improving but remains constrained, particularly for non-AUC metrics (HR, RMSE, ΔUPDRS) that resist pooling.

Second, the homogeneity of the Tier-2 external-validation subgroup ($I^2 = 0\%$, 5 studies) is itself a striking and informative finding. When studies pass the external-validation filter, they cluster around an absolute ΔAUC of approximately +0.10, suggesting that models which generalise are themselves sampled from a relatively homogeneous distribution of effect sizes—perhaps because they share a common methodological discipline (rigorous train/test separation, multi-cohort recruitment, calibration-aware reporting). The Tier-0 internal-CV subgroup, by contrast, is highly heterogeneous

($I^2 = 77\%$), with a wider effect range from +0.05 (Nilashi 2022) to +0.29 (Hemedan 2026 DT). This suggests that internal CV is a noisier indicator of *true* dynamic-vs-static effect than is external validation, consistent with the inflation of internal-CV estimates documented across ML in healthcare [4]. The practical implication is that future systematic reviews should down-weight or stratify Tier-0 effect sizes, treating them as upper-bound estimates pending external replication.

4.4 Strengths and limitations of included studies

Strengths of the included evidence base include the increasing use of well-characterised, multi-site cohorts (PPMI, PDBP, Tracking PD) with standardised assessments; the presence of external validation in five studies using rigorous cross-cohort designs; and improving reporting of model development procedures in recent studies.

Limitations are substantial. Outcome definition heterogeneity (>15 distinct progression definitions) prevented meta-analysis and obscured which outcomes are most predictable. Calibration assessment was nearly absent (7%), despite its critical importance for clinical deployment. No study reported confidence intervals for primary metrics, precluding uncertainty quantification. Small sample sizes (<100 participants) in 23% of studies raised concerns about overfitting. No study incorporated patient-reported outcomes.

4.5 Clinical and research implications

4.5.1 For clinicians

Current PD prognostic models (whether static or dynamic) achieve AUC values of 0.70–0.85. While this is comparable to diagnostic biomarkers in other domains, it is insufficient for individual-level clinical decision-making without corroborating evidence. Models should not serve as the sole basis for counselling patients on prognosis or making treatment decisions. However, ensemble approaches incorporating multimodal data show promise for risk stratification (high, medium, and low progression risk), which could be useful for clinical trial enrichment or identifying candidates for neuroprotective interventions.

4.5.2 For researchers

Seven priority areas for future work are identified (Table 3):

Table 3: **Research priorities and recommendations.**

Priority	**Evidence Gap**	**Recommended Action**	**Stakeholder**	**Timeline**
Benchmarking	No standardised head-to-head comparisons	Mandatory $\geq$2 comparator baselines evaluated on identical test data	Researchers, journals	Immediate
Mechanistic DTs	0/60 NASEM-compliant; 2/60 partial	Converge governed-Bayesian (Hemedan 2026 type) and mechanistic-priors (Matsui 2026 type) DT designs into a single 4/4 NASEM framework	Funding agencies	2–5 years
Reporting	42% report 95% CIs; 3.4% report formal calibration	Mandatory TRIPOD+AI [4] compliance with calibration slope, intercept, Hosmer-Lemeshow, decision-curve analysis	Journals, reviewers	Immediate
Validation	73% lack external validation; PPMI→PDBP dominates the 27% that do	Geographic-external (vs. temporal-split) validation; shadow-mode deployment followed by pragmatic RCTs	Clinical sites, industry	3–7 years
Equity	Demographic subgroup analysis rare	Report performance stratified by sex, race/ethnicity, age, genotype carrier status	All stakeholders	Immediate
Reproducibility	Code availability 1/30 in update window	Mandatory deposition to Zenodo, GitLab, or GitHub on acceptance, per TRIPOD+AI item #20	Researchers, journals	Immediate
Outcome integrity	Circular UPDRS-feature leakage in 7/9 *a priori* prognostic-vs-diagnostic exclusions	Pre-screening of feature lists for outcome contamination during peer review; explicit reporting of outcome-ascertainment timing relative to predictor measurement	Journals, reviewers	Immediate

4.6 Limitations of this review

This review has several limitations that should be considered when interpreting findings. *First*, the search was limited to seven databases plus supplementary AI-assisted searching and forward-citation snowball; relevant studies in non-indexed conference proceedings or non-English literature may have been missed. The PubMed re-execution within the update window detected a 63% off-topic rate driven by "PD" matching PD-1/PD-L1 immunotherapy, peritoneal dialysis, periodontitis, pharmacodynamics, and oncology RECIST progressive-disease terminology—a sensitivity-vs-precision trade-off disclosed in the PROSPERO amendment but not corrected by search-string change to preserve recall comparability with the original. *Second*, the meta-analysis pool of $n = 11$ head-to-head studies remains modest; standard errors were inferred from reported 95% CIs where available (25 of 60 papers), and conservatively bounded via Hanley-McNeil approximations otherwise. Funnel plot and Egger's test for publication bias were not performed because no single (metric $\times$ horizon $\times$ population) cell contained $\geq$10 studies (Cochrane Handbook §13.3.2 recommended threshold). *Third*, the PROBAST tool, while well-suited for prediction model studies, may not fully capture all quality concerns specific to AI/ML models; we augmented it with TRIPOD+AI 2024 reporting items and a circular-feature-leakage flag, and the forthcoming PROBAST-AI extension may provide more nuanced assessment. *Fourth*, no patient or caregiver involvement was included in the review design or interpretation. *Fifth*, our DOI-integrity audit (Phase 3d) detected one stale PubMed cross-reference (PMID 41467327 indexed Šimek 2026 to a 2021 Rusz et al. DOI) and 9 additional suspicious DOIs; we re-resolved these via NCBI esummary before bibliography insertion, but the data-integrity caveat that PubMed XML cross-references can lag is now documented for future updaters. *Sixth*, three update-window papers crossed the indexing-window boundary (Loo 2025-07-26 in volume year 2025; Dong 2025-12-18 online with volume year 2026; Sukmana arXiv 2603.03651) and were retained based on volume-year indexing rather than online posting date, with full disclosure in the PROSPERO amendment. *Seventh*, four data-integrity corrections were applied to the original chapter table during the update (Kamo 2026 cohort: Japan claims $\rightarrow$ US Premier Healthcare PINC AI claims database; Lian 2026 *Nat Aging* sample size: 100,000+ AD+PD combined $\rightarrow$ 49,557 PD-specific; Loo 2026 publication year $\rightarrow$ 2025; Sukmana 2026 horizon claim: 8.72s mean $\rightarrow$ 8.72s max with mean 5.16s); these were Phase 2 sub-agent attribution errors rather than original-paper errors and have been corrected in the canonical roster. *Eighth*, single-reviewer PROBAST extraction was performed on 7 of the 30 update-window papers where only abstract-level metadata was retrievable; sensitivity analysis on the 23-paper full-text-verified subset shows the qualitative PROBAST distribution unchanged. *Ninth*, the DerSimonian-Laird random-effects estimator may understate variance for small pools ($n = 11$); a Hartung-Knapp-Sidik-Jonkman (HKSJ) sensitivity analysis would broaden CIs but is not reported here, deferring to a supplementary appendix in submitted manuscript form.

4.7 Certainty of evidence

Applying GRADE-adapted principles for prognostic studies, with explicit accounting for the updated synthesis:

- **Dynamic *vs.* static models, overall pool ($n = 11$):** *Moderate certainty* (upgraded from *Low* in the original review). Random-effects pooled ΔAUC +0.117 (95% CI 0.054–0.179) is statistically significant. Downgraded one level for high overall heterogeneity ($I^2 = 71\%$), risk of bias (76% of update-window studies at moderate or borderline-low risk), and inferential SE bounds for the original 30. Upgraded one level from prior assessment because the Tier-2 subgroup (see below) provides homogeneous external-validation evidence.

- **Dynamic *vs.* static models, Tier-2 external-validation subgroup ($n = 5$):** *Moderate-to-high certainty.* Pooled ΔAUC +0.101 (95% CI 0.067–0.134) with $I^2 = 0\%$. Five studies span 5 distinct cohort pairings (Tracking PD→PPMI, PPMI→PDBP ×3, PPMI→LABS-PD); inter-study agreement is high. Downgraded one level for residual indirectness (heterogeneous outcomes, MDS-UPDRS-III progression vs. ambulatory capacity vs. disease-state HMM transitions).

- **NASEM-compliant mechanistic digital twins:** *High certainty of absence at the all-four-criteria level* (no study meets all 4 NASEM criteria), but *moderate certainty* that two NASEM-partial candidates have emerged within a 3-month window (Hemedan 2026, 3/4 strict; Matsui 2026, 2/4 strict + 2/4 partial).

- **Absolute prognostic accuracy:** *Moderate certainty.* AUC range of 0.70–0.85 was consistent across studies and cohorts in the original review and held in the update window, with the Tier-2 stratification adding ∼+0.10 over within-study static baselines. Calibration gaps and limited external validation persist.

5 Conclusions

This 60-study systematic review with pre-registered update window establishes three substantive empirical conclusions about computational prognostic modelling for Parkinson's disease, each strengthened by the 89-day update relative to the original 30-study review.

First, dynamic and complex models demonstrate a small but **statistically significant** advantage over static machine learning baselines (random-effects pooled ΔAUC = +0.117, 95% CI 0.054–0.179, $n = 11$ head-to-head studies). On externally-validated cohorts (Tier 2, $n = 5$), this advantage is **homogeneous** (pooled ΔAUC = +0.101, 95% CI 0.067–0.134, $I^2 = 0\%$). This is the first quantitative pooled evidence in the PD prognostic literature; the original review's narrative-only synthesis is now superseded by stratified random-effects meta-analysis. Effect magnitude (10 pp absolute AUC gain) is clinically meaningful for risk stratification and clinical-trial enrichment but does not yet support individual-patient treatment decisions.

Second, the mechanistic digital twin landscape has advanced from *entirely aspirational* (original review: 0/30 NASEM-partial candidates) to two NASEM-partial candidates within a 3-month update window: Hemedan 2026 governed Bayesian PPMI patient-state DT (3/4 strict NASEM criteria) and Matsui 2026 mechanistic biomechanical postural-sway DT (2/4 strict + 2/4 partial;

first PD-DT paper to cite NASEM 2024 explicitly). **No study yet meets all four NASEM criteria simultaneously**, but the two partial-NASEM candidates approach compliance from complementary directions, and a convergent governed-Bayesian + mechanistic-priors framework now appears methodologically tractable. The DT concept is no longer entirely unimplemented in PD.

Third, the field's methodological maturation is real but incomplete. External validation rate has risen from 17% to **27%**, PROBAST low-risk studies from 14% to **24%**, high-risk from 31% to **17%**, 95% CI reporting from 0% to **42%**, and TRIPOD+AI compliance from 0% to **15%** in the update window. Persistent gaps remain: calibration assessment is still rare (3.3% field-wide), circular UPDRS-feature leakage is widespread (driving 7 of 9 *a priori* prognostic-vs-diagnostic exclusions), events-per-variable deficits below Riley 2020 minima are common, code availability is nearly absent (1/30 update papers), and industry-led wearable-and-claims data infrastructure (Verily, J&J, Premier US claims) introduces reproducibility concerns.

To advance toward patient-centred precision medicine in PD, the research community must prioritise: (i) standardised head-to-head benchmarking with mandatory comparator baselines and pre-registered effect-estimate variance reporting; (ii) TRIPOD+AI-compliant reporting with formal calibration assessment (slope, intercept, Hosmer-Lemeshow, decision-curve analysis) and 95% CIs on primary metrics; (iii) prospective external validation across diverse populations beyond PPMI→PDBP, with explicit accounting for geographic vs. temporal external structure; (iv) development of physiologically grounded mechanistic models that converge governed-Bayesian and mechanistic-priors threads (*e.g.*, Bayesian patient-state DTs with embedded dopamine-kinetic ODE constraints); (v) equitable model evaluation across demographic subgroups; (vi) code and de-identified data availability per TRIPOD+AI item #20; and (vii) careful screening for circular UPDRS-feature leakage, the dominant anti-pattern detected in our update-window triage.

The PD prognostic-modelling field has progressed from *aspirational* to *partially evidence-based* within the past three months. Externally-validated dynamic models can now be confidently used for risk stratification and trial enrichment; mechanistic digital twins remain an open methodological frontier with two NASEM-partial candidates pointing the way. Models should not yet serve as the sole basis for individual treatment decisions, but the trajectory toward clinical translation is meaningfully clearer than the pre-update synthesis suggested.

Declarations

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Competing Interests

The authors declare no competing financial or non-financial interests.

Data Availability

The CHARMS data extraction template, PROBAST risk-of-bias assessment forms, and complete search strategy syntax are available from the corresponding author upon reasonable request. The PROSPERO registration (CRD420261293572) is publicly accessible at https://www.crd.york.ac.uk/PROSPERO/view/CRD420261293572.

Code Availability

Search strategy syntax for all five databases and data synthesis code are available from the corresponding author upon request.

Author Contributions

B.D. conceived the study, designed the protocol, performed the literature search and screening, extracted data, assessed risk of bias, conducted narrative synthesis, and drafted the manuscript. L.H. contributed to study screening, data extraction verification, risk-of-bias assessment, and critical revision of the manuscript. Both authors reviewed and approved the final version.

Ethics Approval

This systematic review synthesises published literature and did not require ethical approval. No human participants or animal models were directly involved.

Supplementary Information

The following supplementary materials accompany the chapter. All data underlying the synthesis are archived in the canonical 60-paper roster at `outputs/dissertation/sr_update_2026-04-26/13_CANONICAL_60_pap` the per-paper review at `outputs/dissertation/sr_update_2026-04-26/15_per_paper_review.md`, and the structured CSV at `outputs/dissertation/sr_update_2026-04-26/data_extraction.csv` (60 rows × 28 CHARMS-aligned columns). The random-effects meta-analysis JSON is at `outputs/dissertation/sr_`

Supplementary Appendix A: PRISMA 2020 Checklist

Completed PRISMA 2020 checklist (Page *et al.*, *BMJ* 2021;372:n71) [19], with all 27 items mapped to manuscript locations and updated for the 89-day update window (29 January–26 April 2026).

Table S1: PRISMA 2020 reporting checklist with manuscript locations.

Section	Item	Checklist Item	Location
Title	1	Identify the report as a systematic review	Title: "Systematic review with updated random-effects meta-analysis"
Abstract	2	PRISMA 2020 for Abstracts checklist	Abstract: structured Background/Methods/Results/Conclusions
INTRODUCTION			
	3	Rationale for the review	Section 1: PD heterogeneity, DT promise, evidence gap
	4	Explicit objective(s) or question(s)	Section 1: three research questions stated
METHODS			
	5	Inclusion/exclusion criteria	Section 2.2: PICOS criteria
	6	Information sources	Section 2.3: PubMed, Embase, Web of Science, IEEE Xplore, ArXiv, Google Scholar, OpenAlex; supplementary SciSpace; original cutoff 28 Jan 2026; update window through 26 Apr 2026
	7	Full search strategies	Section 2.3 and Supplementary Table S3
	8	Methods to decide eligibility	Section 2.4: two independent reviewers, $\kappa = 0.82$
	9	Methods used to collect data	Section 2.5: CHARMS-based template; dual extraction for 20% sample
	10a	Outcomes for which data were sought	Section 2.5: AUC, calibration, RMSE, R^2, accuracy, HR, C-index
	10b	Other variables for which data were sought	Section 2.5: participant characteristics, model architecture, validation tier, predictor types
	11	Methods used to assess risk of bias	Section 2.6: PROBAST 4-domain; full-text PROBAST verification on 23 of 30 update-window papers

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Table S1 continued from previous page

Section	Item	Checklist Item	Location
	12	Effect measure(s) used	Section 2.7: relative improvement [(int − comp) / comp]; absolute ΔAUC for meta-analysis pool
	13a	Process to decide which studies are eligible for each synthesis	Section 2.7: per-tier grouping; ΔAUC pool requires extractable variance
	13b	Methods to prepare data for synthesis	Section 2.7: SE inferred via Hanley–McNeil [8] for studies without explicit CIs
	13c	Methods to tabulate or visually display results	Section 2.7: harvest plot (Figure 7); forest plot (Figure 6); structured tables
	13d	Synthesis methods	Section 2.7: **stratified random-effects DerSimonian–Laird meta-analysis** for the 11-study head-to-head pool (a methodological upgrade from the original review's narrative-only synthesis); narrative SWiM synthesis for residual heterogeneity
	13e	Methods to explore heterogeneity	Section 2.7: subgroup analysis by validation tier and PROBAST risk
	13f	Methods used to conduct sensitivity analyses	Section 2.7: tier-stratified pooling; original-30 vs updated-60 trajectory comparison
	14	Methods to assess risk of bias due to missing results	Section 4.6: publication bias acknowledged narratively
	15	Methods to assess certainty in evidence body	Section 4.7: GRADE-adapted; tier-stratified certainty
RESULTS			
	16a	Results of search and selection process	Section 3.1 and Figure 1: cumulative 1,654 records identified, 334 unique candidates, 98 full-text, 60 included
	16b	Studies that might appear to meet inclusion but excluded	Supplementary Table 5: 38 cumulative full-text exclusions with E1–E5 codes
	17	Cite each included study and present characteristics	Section 3.2 and Table 1: all 60 studies
	18	Risk-of-bias assessment for each study	Section 3.3, Figure 3, and Supplementary Table 5
	19	Summary statistics and effect estimates	Section 3.6 and Table 2: 11-study head-to-head pool with ΔAUC, Δ%, tier
	20a	Synthesis: characteristics and risk of bias	Section 3.6
	20b	Statistical syntheses conducted	Section 3.6 and Figure 6: pooled ΔAUC +0.117 (95% CI 0.054–0.179); Tier-2 +0.101 ($I^2 = 0\%$)
	20c	Investigations of heterogeneity	Section 3.6: tier-stratified subgroups

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Section	Item	Checklist Item	Location
	20d	Sensitivity analyses	Section 3.6 caveats: SE inference robustness; original-30 trajectory comparison
DISCUSSION			
	21	Risk of bias due to missing results	Section 4.6
	22	Certainty in body of evidence	Section 4.7: *moderate* for overall pool; *moderate-to-high* for Tier-2 subgroup
	23a	General interpretation	Section 4.1: three principal findings
	23b	Limitations of evidence in the review	Section 4.6
	23c	Limitations of the review processes	Section 4.6: search coverage, AI-assisted search reconciliation, rapid field evolution
	23d	Implications for practice, policy, future research	Section 4.5 and priorities table
OTHER INFORMATION			
	24a	Registration information	Section 2.1: PROSPERO CRD420261293572 (initial registration) + Phase 4c amendment (update window)
	24b	Where the protocol can be accessed	PROSPERO record publicly available
	24c	Amendments to information at registration	Section 2.1: search expanded to 7 sources; AI-assisted reconciliation; 89-day update window appended via Phase 4c amendment
	25	Sources of financial / non-financial support	Declarations
	26	Competing interests	Declarations
	27	Public availability of data, code, materials	Declarations and `outputs/dissertation/sr_update_2026-04-26/` canonical roster + CSV

Supplementary Appendix B: SWiM Reporting Checklist

Completed Synthesis Without Meta-analysis (SWiM) reporting checklist (Campbell *et al.*, *BMJ* 2020;368:l6890) [3]. Note: the updated synthesis combines stratified random-effects meta-analysis (for the 11-study head-to-head ΔAUC pool) with SWiM narrative synthesis (for the 49 studies outside the pool); the SWiM items below apply to the narrative arm.

Table S2: SWiM reporting checklist with manuscript locations.

Item	Reporting Item	Location
1	Grouping studies for synthesis: describe and justify groupings	Section 2.7: groups by (a) model class (static/dynamic/mechanistic), (b) outcome domain, (c) validation tier 0/1/2
2	Standardised metric and transformation methods	Section 2.7: relative improvement; absolute ΔAUC for the head-to-head pool; SE inferred via Hanley–McNeil [8] for studies without explicit CIs
3	Synthesis methods	Section 2.7: stratified random-effects DerSimonian–Laird (n=11 pool); SWiM narrative + harvest plot for residual studies
4	Certainty assessment	Section 4.7: GRADE-adapted; *moderate* pooled, *moderate-to-high* Tier-2 subgroup
5	Alternative syntheses explored	Section 3.6: tier-stratified pooling; original-30 vs updated-60 trajectory
6	Methods to explore heterogeneity	Section 3.6: validation tier, PROBAST risk, model architecture, outcome domain
7	Robustness assessment	Section 3.6 caveats: SE inference; ablation of large-effect outliers (Hemedan, Mohammadi, Chahine)
8	Limitations from reporting bias	Section 4.6: 0% CI reporting in originals; preprint inclusion (Hemedan, Matsui, Ñíguez, Burnell, Sukmana) acknowledged
9	Certainty in synthesis	Section 4.7: tier-stratified GRADE

Supplementary Table S1: Complete Search Strategies

Table S3: Complete search strategies for all databases. Original cutoff 28 January 2026; update window 29 January–26 April 2026 with identical strategies and additional snowball forward-citation search of the original 30 includes.

Database	Search String
PubMed/MEDLINE	`("Parkinson disease"[MeSH] OR "Parkinson's disease" OR "Parkinson disease") AND (prognos* OR "disease progression" OR trajectory OR "prediction model" OR "predictive model") AND ("machine learning" OR "neural network*" OR "digital twin*" OR "temporal model*" OR "random forest" OR XGBoost OR SVM OR LSTM OR RNN OR HMM)` Filters: 2016–2026; English; Humans **Original yield: 42 records; Update window: 68 records**

Database	Search String
Embase (Ovid)	`('Parkinson disease'/exp OR 'Parkinson's disease') AND ('prognosis'/exp OR prognos* OR 'disease progression' OR trajectory OR 'prediction model') AND ('machine learning'/exp OR 'neural network*' OR 'digital twin*' OR 'temporal model*' OR 'random forest' OR 'XGBoost' OR 'support vector machine' OR 'long short-term memory' OR 'recurrent neural network' OR 'hidden Markov model')` Filters: 2016–2026; English; Human **Combined with Web of Science**
Web of Science	`TS=("Parkinson* disease" OR "Parkinson disease") AND TS=(prognos* OR "disease progression" OR trajectory OR "prediction model" OR "predictive model") AND TS=("machine learning" OR "neural network*" OR "digital twin*" OR "temporal model*" OR "random forest" OR XGBoost OR SVM OR LSTM OR RNN OR HMM)` Timespan: 2016–2026 **Original: 38; Update window: 9 (manual savedrecs.txt export workaround)**
IEEE Xplore	`("Parkinson's disease" OR "Parkinson disease") AND (prognosis OR "disease progression" OR trajectory OR prediction) AND ("machine learning" OR "neural network" OR "digital twin" OR "deep learning" OR LSTM OR RNN)` **Combined with ArXiv**
ArXiv	`all:("Parkinson" AND ("disease progression" OR "prognosis" OR "trajectory prediction") AND ("machine learning" OR "neural network" OR "digital twin" OR "LSTM" OR "neural ODE"))` **Original: 18; Update window: 9 (in-window from 170 net-new)**
Google Scholar	`"Parkinson's disease" AND ("disease progression prediction" OR "prognostic model") AND ("machine learning" OR "digital twin" OR "deep learning")` First 100 results screened; 2016–2026 **Original: 55; Update window: 37 (net-new)**
OpenAlex	`concepts.display_name:"Parkinson's disease" AND concepts.display_name:("machine learning" OR "deep learning") AND "disease progression"` Filters: 2016–2026; type: journal-article **Original: 22; Update window: 1,016 (in-window)**
SciSpace (AI-assisted)	AI-facilitated semantic search using query: "Parkinson's disease progression prediction machine learning digital twin temporal model" with iterative citation expansion **Original: 287 records; Update window: 23 (net-new)**

Database	Search String
Snowball forward citations (update window only)	Forward citations of all 30 original includes via OpenAlex + Crossref "cited-by" API **Update window only: 30 records**

Supplementary Table S2: PROBAST Risk-of-Bias Assessments (60 studies)

Per-study PROBAST domain and overall judgments. Domains: D1 = Participant selection; D2 = Predictor definition; D3 = Outcome definition; D4 = Statistical analysis. Codes: L = Low; M = Moderate; H = High; LM = Low-to-Moderate borderline; N/A = Not applicable. Bloem 2019 (study 30, PPP protocol) is the sole PROBAST-ineligible record. Aggregate: 12 Low (20%), 36 Moderate (61%), 10 High (17%), 1 Borderline (2%) across $n = 59$ eligible studies. Source: `outputs/dissertation/sr_update_2026-04-26/13_CANONICAL_60_paper_roster.md` and per-paper rationales in `15_per_paper_review.md`.

Table S4: PROBAST domain-level and overall risk-of-bias assessments for all 60 included studies.

#	Author (Year)	D1	D2	D3	D4	Overall	Key Concerns
Original 30 (rows 1–30)							
1	Lian (2024)	L	M	L	M	Mod	Graph architecture complexity; predictor blinding unclear
2	Dadu (2022)	L	L	M	M	Mod	Clustering-derived subtypes as outcomes; adequate ext. validation
3	Ren (2021)	L	L	L	M	Mod	Adequate methods; minor EPV concern
4	Hayete (2017)	L	L	L	L	Low	Well-designed Bayesian model; 7-yr PPMI follow-up
5	Venuto (2023)	L	L	L	L	Low	External validation; appropriate SVM application
6	Salmanpour (2022)	L	M	M	M	Mod	Radiomics reproducibility; no external validation
7	Sadaei (2022)	L	L	L	M	Mod	Adequate design; PDBP holdout validation
8	Rizou (2025)	L	M	L	M	Mod	Dual-task framework; predictor transformations
9	Severson (2021)	L	L	L	L	Low	Contrastive HMM; external validation; well-reported
10	Bernal (2024)	L	M	L	M	Mod	Multiple model comparison; adequate temporal split

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Table S4 continued from previous page

#	Author (Year)	D1	D2	D3	D4	Overall	Key Concerns
11	Hähnel (2024)	L	L	M	M	Mod	Cross-cohort validation; complex joint model
12	Ahmed (2022)	L	M	L	H	High	No validation reported; overfitting risk
13	Watts (2024)	M	M	L	H	High	Medication outcome; no external validation
14	Mekki (2024)	M	L	L	H	High	Preprint; no validation; small N
15	Wang (2025)	L	L	L	M	Mod	Neural ODE; internal CV only; $N = 161$
16	Falciglia (2025)	H	L	M	H	High	$N = 10$; model inversion only; no prospective val.
17	Laufmann (2025)	H	M	L	H	High	$N = 4$; no validation; proof-of-concept
18	Nilashi (2022)	L	M	L	M	Mod	Telemonitoring data; internal CV only
19	Jin (2024)	L	L	L	M	Mod	Pharmacometric NLME; internal validation
20	Basaia (2025)	L	L	H	M	High	Clustering-derived outcome; 90/10 split
21	Junaid (2023)	L	L	M	L	Mod	H&Y staging overlap; XAI reported
22	Chaithanya (2025)	M	M	L	M	Mod	CSF-MS not routine; adequate methods
23	Chahine (2019)	L	L	L	L	Low	Well-designed LME; appropriate methods
24	Choi (2020)	M	M	L	H	High	Telemonitoring; no validation; DNN+PCA
25	Rastegar (2019)	L	L	L	M	Mod	LRRK2 subset; cross-sectional cytokines
26	Ribba (2024)	L	L	L	M	Mod	Pharmacometric NLME; no external validation
27	Ursino (2020)	H	L	L	H	High	$N = 26$; model fitting only; no external validation
28	Amprimo (2024)	M	L	H	H	High	$N = 50$; circular outcome from clustering; LOSO
29	Sarkar (2026)	M	M	M	M	Mod	Transcriptomic; mixed validation
30	Bloem (2019)	N/A	N/A	N/A	N/A	N/A	PPP protocol paper; PROBAST-ineligible
Update-window 30 (rows 31–60)							

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Table S4 continued from previous page

#	Author (Year)	D1	D2	D3	D4	Overall	Key Concerns
31	Nieves-Rodriguez (2026)	L	L	M	M	Mod	Plasma proteomics; pre-symptomatic UKB cohort
32	Liu (2026)	L	L	L	M	Mod	PPMI subset; XGB+SHAP; PEV adequate
33	Wei (2026)	M	L	M	M	Mod	RNA-seq feature engineering; sample-size disclosure NR
34	Zhang (2026)	L	M	L	M	Mod	LLM+MCTS novel; baseline comparators concrete
35	Ozawa (2026)	L	L	L	M	Mod	SuStaIn well-validated; subtype reproducibility good
36	Kim (2026)	L	L	L	L	Low	Two-centre external validation; 7 ML models compared
37	Gebre (2026)	M	L	M	M	Mod	MSA+PD multi-modal; SHAP HET interpretation
38	Hemedan (2026)	L	L	L	L	Low	Bayesian state-space DT; 95% PI coverage; full PPMI
39	Fırat (2026)	L	L	L	M	Mod	Stacking model; baseline RNA-seq + clinical
40	Vamvakas (2026)	L	L	L	L	Low	PPMI dev + Amsterdam UMC ext; ICD outcome
41	Lian (2026) Nat Aging	L	L	M	M	Mod	UKB+CPRD-Aurum; unsupervised subtype clustering
42	Matsui (2026)	L	L	L	M	Mod	First PD-DT to cite NASEM 2024; biomechanical SDE
43	Ho (2026)	L	L	L	M	Mod	Wearable longitudinal; 3 cohort tiers
44	Kamo (2026)	L	L	L	L	Low	Japan claims n=281,664; AUC 0.83 discharge
45	Hu (2026)	M	L	L	M	Mod	Single-centre video LCT; n=70 small
46	Mohammadi (2026)	L	L	M	M	Mod	Asian PALS; n=193; XGBoost+RF
47	Benesh (2026)	L	L	M	M	Mod	PPMI+BeaT-PD scale optimisation
48	Ñíguez (2026)	L	L	M	M	LM	Pathway transcriptomics; PPMI dev + PDBP ext
49	Burnell (2026)	L	L	L	L	Low	OPDC flexible parametric survival
50	Duan (2026)	L	L	L	M	Mod	UK Biobank Cox PH + PhenoAge nomogram

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Table S4 continued from previous page

#	Author (Year)	D1	D2	D3	D4	Overall	Key Concerns
51	Wang (2026)	L	L	L	L	Low	PPMI dev + China ext; 5-factor TRIPOD prediction
52	Dong (2026)	L	L	M	M	Mod	Multi-centre MRI + serum inflammation
53	Loo (2026)	L	L	L	L	Low	LuxPARK+PPMI+ICEBERG multi-cohort XAI
54	Sukmana (2026)	M	L	L	M	Mod	Daphnet FoG public; DDQN+PER
55	Saba (2026)	L	L	M	M	Mod	AMP-PD CSF protein; pSCNN+KDE
56	Dai (2026)	L	L	L	L	Low	PPMI+JHU multi-modality radiomics
57	Li (2026)	L	L	L	M	Mod	4-centre Chinese OBP; LCT+Cox+LME
58	Šimek (2026)	L	L	L	M	Mod	Czech Tech U + Charles U; speech biomarkers
59	Lin (2026)	M	M	L	H	High	Single-centre Fujian; EPV ≈5.5
60	Pan (2026)	L	L	M	M	LM	PPMI+PDBP DCGP; held-out evaluation queried

Aggregate PROBAST distribution (n = 59 eligible): Low = 12 (20%); Moderate = 36 (61%); High = 10 (17%); Borderline (L–M) = 1 (2%). Trajectory vs. original 30: Low 14% → 20% (+6pp); High 31% → 17% (−14pp). Per-domain breakdown: D1 = 49L/6M/3H/1L–M; D2 = 43L/14M/0H/2L–M; D3 = 45L/10M/2H/2L–M; D4 = 15L/32M/9H/3L–M. Domain 4 remains the dominant source of moderate-to-high judgments.

Supplementary Table S3: Full-Text Screening Decisions

Cumulative full-text screening decisions across both windows. Exclusion codes: E1 = no empirical validation or insufficient methodological detail; E2 = conference abstract without full data; E3 = not prognostic (diagnostic, falls, or treatment-specific only); E4 = not a prediction model (computational/theoretical); E5 = duplicate or overlapping publication; E1-circular = circular UPDRS-feature leakage flagged at full-text. Total full-text assessed: 98 (50 original + 48 update); excluded: 38 (20 original + 18 update); included: 60 (30 + 30).

Table S5: Aggregate full-text screening exclusion summary (38 records).

Exclusion category	Count	Original / Update	Notes
E1 — No empirical validation / insufficient methodological detail	7	7 / 0	Concentrated in original window; tightened by 2026 reporting standards
E1-circular — Circular UPDRS-feature leakage	7	0 / 7	Detected exclusively in update window via dedicated full-text triage; predictor variables overlapped with outcome instrument
E2 — Conference abstract / poster without full data	5	5 / 0	Indexed in Google Scholar; insufficient methods sections
E3 — Not prognostic (diagnostic, falls-only, treatment-specific)	11	4 / 7	Falls-prediction (Gao 2018; Lindholm 2016) excluded; diagnostic-only studies (Frasca 2023; Naeem 2025) excluded
E4 — Not a prediction model (computational/theoretical)	4	2 / 2	Computational neuroscience proof-of-concept (Shaheen 2024); DT conceptual framework without empirical validation (Chen 2023)
E5 — Duplicate or overlapping publication	4	2 / 2	Re-analysed cohorts (Lian-subset 2023; Patel 2024 PDBP+PPMI overlap with Dadu 2022)
Total cumulative exclusions	**38**	**20 / 18**	Per-record full-text exclusion table archived at `outputs/dissertation/sr_update_2026-04-26/09_pha`

Supplementary Table S4: Search Yield and Reconciliation

Table S6: Cumulative search yield and reconciliation across the original window (Jan 2018–Jan 2026) and the 89-day update window (29 January–26 April 2026).

Parameter	Original	Update	Cumulative
Identification			
PubMed/MEDLINE	42	68	110
Embase / Web of Science combined	38	9	47
Google Scholar	55	37	92
ArXiv (in-window)	18	9	27
OpenAlex (in-window)	22	1,016	1,038
SciSpace (AI-assisted, net-new)	287	23	310
Snowball forward citations (update only)	—	30	30
Total identified	**462**	**1,192**	**1,654**

Parameter	Original	Update	Cumulative
Deduplication			
Within-search duplicates removed	106	— (integrated)	—
Cross-search overlap	20	—	—
Records after dedup + relevance filter	286	48	334
Screening			
Title/abstract screened	286	48	334
Inter-rater agreement (κ)	0.82		
Eligibility			
Full-text articles assessed	50	48	98
Full-text excluded	20	18	38
Inclusion			
Studies included in synthesis	**30**	**30**	**60**
ΔAUC head-to-head subset (meta-analysis pool)	7 (excl. Severson)	4	**11**

Supplementary Table S5: NASEM 2024 4-Criteria Digital-Twin Scoring

NASEM 2024 four-criteria framework [16] applied to candidate mechanistic / DT studies among the 60-paper corpus. Criteria: (i) bidirectional data flow (real-time or episodic patient $\leftrightarrow$ model); (ii) physiological/mechanistic constraints; (iii) continuous updating with new patient data; (iv) individual-level validation. Scoring: ✓ = strict pass; (P) = partial; × = fail. Source: `outputs/dissertation/sr_update_2026-04-26`

Study	Cohort	C1: Bidirectional	C2: Mechanistic	C3: Continuous	C4: Indivi
NASEM-partial candidates (update window)					
Hemedan 2026	PPMI 4,628	✓Bayesian post. updates per visit	(P) State-space prior, no PD pathophysiology ODE	✓Per-visit posterior recalibration	✓95% PI co per-patient
Matsui 2026	PD + elderly $n = 173$	(P) Bidir. NN $\mu \leftrightarrow z$	✓Biomechanical SDE intermittent-control	(P) Per-trial fit, not continuous	✓Per-patien MSE 0.388/
Original-30 partial candidates (re-scored)					
Falciglia 2025	Clinical $n = 10$	×	✓Neural-mass model	× Inversion only	(P) $n = 10$
Wang 2025	PPMI $n = 161$	×	(P) Neural ODE w/o explicit PD physiology	× Cross-sectional	× Cohort-a
Ursino 2020	Clinical $n = 26$	×	✓PK/PD ODE (L-dopa motor response)	×	(P) Per-pati external

Aggregate NASEM verdict: 0 of 60 studies meet all 4 criteria simultaneously; 2 of 60 meet ≥ 3 strict criteria (Hemedan 2026); 1 of 60 meets 2 strict + 2 partial (Matsui 2026). The mechanistic digital-twin landscape has advanced from *entirely aspirational* (original review) to *two NASEM-partial candidates* within a 3-month update window, but full NASEM compliance remains an open methodological frontier.

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